CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL

JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

BEST FROM KOTA LECTURES <<< CLICK

QUIZES <<< CLICK

REVISION FLASH CARDS &

MANY MORE <<< CLICK

Classification and Nomenclature, Isomerism & General Organic Chemistry

Direction : Q.1 To Q.25
Statement are true or false

1. Hybridization influences the bond length and bond enthalpy

CCN185

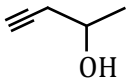
2. In $CH_3–CH=CH–CH_3$ all the atoms are in same plane because formation of π bond is possible if p orbitals on adjacent atoms are parallel.

CCN186

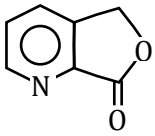
3. Number of σ bond and π bond in $CH_3–CH=CH–CN$ are 9 and 3.

CCN187

4. Bond line notation of pent-4-yn-2-ol is

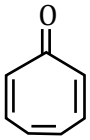


CCN188

5. Number of hetero atom in 

, are 3

CCN189

6. 

is Benzenoid structure

CCN190

7. 

(THF) is Alicyclic compound

CCN191

8. Pyridine, Furane and Thiophene all are Heterocyclic aromatic compound

CCN192

9. Ethanoic acid and Butanoic acid are Homologues and can be represented by same general molecular formula.

CCN193

10. Number of alkyl groups obtained from $(CH_3)_3C–CH(CH_3)_2$ are 4

CCN194

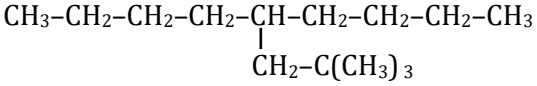
11. The correct name of 6-ethyl-3-methyl octane is 3-ethyl-6-methyl octane

CCN195

12. The name of branched chain alkyl group and their position are placed in parenthesis (in bracket) in IUPAC name

CCN196

13. The IUPAC name of

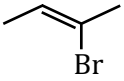


is 5-(2,2-Dimethyl ethyl) nonane

CCN197

14. If there are two different F.G. in a compound than one, which is principle functional is named as prefix and another is substituent which is used as suffix.

CCN198

15. IUPAC name of 

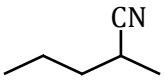
is

3-Bromobut-2-ene

CCN199

16. $CH≡C–CH=CH–CH_2–CH=CH_2$ is named as hepta-3,6-dien-1-yne

CCN200

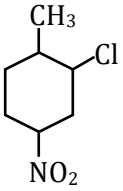
17. 

is named as 2-Cyanopentane

CCN201

18. But-2-en-3-amine is correct IUPAC name

CCN202

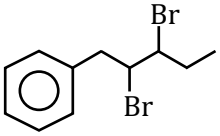
19. 

is named as

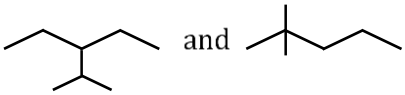
1-chloro-2-methyl-5-nitro cyclohexane

CCN203

20. The correct structure of 2,3-dibromo-1-phenyl pentane is



CCN204

21. 

are positional isomers

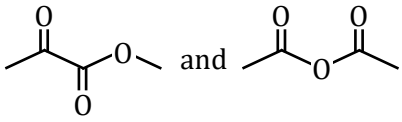
CIS131

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

22. $C_4H_{10}O$ can show metamerism

CIS132

23.  and

are functional isomers

CIS133

24. Heterolytic cleavage of $CH_3–Cu$ gives $\overset{\ominus}{C}H_3$ and $\overset{\oplus}{C}u$

CGC265

25. In $CH_3–I$ carbon is electrophilic centre

CGC266

26. The incorrect statement about Inductive effect ?

(1) In chloroethane ($CH_3–CH_2–Cl$) the polar C–Cl bond induces polarity in the adjacent bond.

(2) The effect decreases rapidity as the number of intervening bond increases.

(3) This effect becomes vanishingly small after three bonds.

(4) Aryl is electron donating groups (+I)

CGC267

27. Which is incorrect order regarding polar bond ?

(1) $CH_3–Br > CH_3–H$

(2) $CH_3–OH > CH_3–NH_2$

(3) $CH_3–CH_3 > CH_3–OH$

(4) $CH_3–F > CH_3–Cl$

CGC268

28. The incorrect statement about resonance effect?

(1) Canonical structures contribute to the actual structure in proportion to their stability.

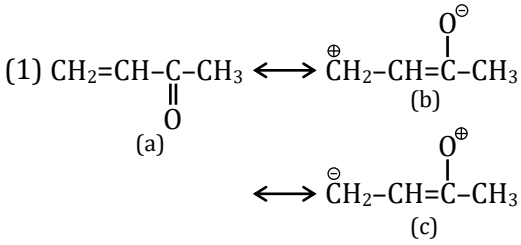
(2) The energy of actual structure is lower than that of any of the canonical structure.

(3) The difference in energy between the highest energy canonical structure and the lowest energy canonical structure is called as resonance energy.

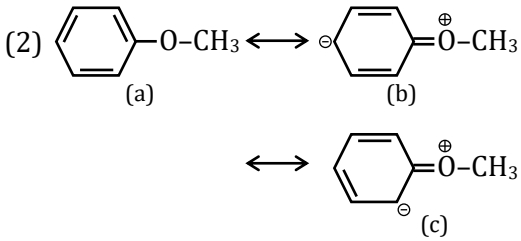
(4) The more number of canonical structure, the more is the resonance energy.

CGC269

29. Give stability order of following resonating structures

(1) 

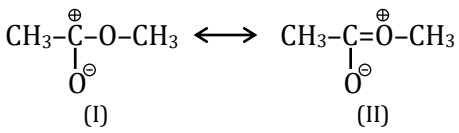
(a) (b) (c)

(2) 

(a) (b) (c)

CGC270

30. Why following two structures are not the major contributions to the real structure of $CH_3–COOCH_3$.



(I) (II)

CGC271

31. Which is ⊖R group ?

(1) $–O–\overset{\overset{O}{\|}}{C}–R$ (2) $–NH–\overset{\overset{O}{\|}}{C}–R$

(3) –Br (4) $–\overset{\overset{O}{\|}}{C}–O–R$

CGC272

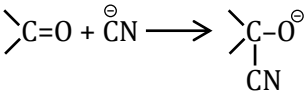
32. Correct statement about electrometric effect?

(1) Diethyl ether shows E-effect

(2) The effect is annulled as soon as the attacking reagent is removed

(3) The inductive effect predominates over E-effect

(4) The following is an example of +E-effect



CGC273

AIM : FREE EDUCATION TO ALL ♥ APUL

33. Incorrect statement about H-effect is -

(1) H-effect is permanent effect

(2) σ electron of C–H bond enter into the partial conjugation with empty p orbital.

(3) This type of overlapping destabilise the carbocation.

(4) Delocalization of electrons by hyperconjugation is also possible in alkene.

CGC274

34. $CH_3-\overset{\oplus}{C}H_2$ is more stable than $\overset{\oplus}{C}H_3$. It is explained by, except-

(1) +I of $-CH_3$ increases stability of $CH_3-\overset{\oplus}{C}H_2$

(2) Hyperconjugation interaction in $CH_3-\overset{\oplus}{C}H_2$

(3) The C–H σ bond helps in dispersing the positive charge.

(4) Electrons of C–H bond can be delocalized into empty 3p orbital.

CGC275

35. Incorrect statement about $CH_3-O-CH_2-CH_2-CH_3$ is

(1) It is unsymmetrical ether

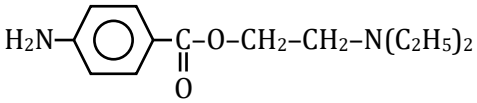
(2) It is IIIrd member of ether

(3) It is IInd member of unsymmetrical ether

(4) It is IIIrd member of unsymmetrical ether

CGC276

36. Which functional group is not present in



(1) 1° Amine (2) 2° Amine

(3) 3° Amine (4) ester

CGC277

37. Why alkyl group act as electron donor when attached to π system

(1) Due to +I effect of alkyl group

(2) Due to hyperconjugation effect

(3) (1) and (2) both

(4) Due to +M of alkyl group

CGC278

38. Which is an example of addition reaction ?

(1) $CH_3-CH_2-Br + \overset{\ominus}{H}S \rightarrow CH_3-CH_2-SH$

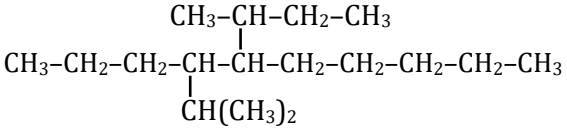
(2) $(CH_3)_2C=CH_2 + H-Cl \rightarrow (CH_3)_2\underset{\underset{Cl}{|}}{C}-CH_3$

(3) $CH_3-CH_2-Br + \overset{\ominus}{O}H \rightarrow CH_2=CH_2 + H_2O + \overset{\ominus}{Br}$

(4) $(CH_3)_3C-CH_2-OH + H-Br \rightarrow (CH_3)_2CBr\,CH_2-CH_2-CH_3 + H_2O$

CGC279

39. The correct IUPAC name of



(1) 5-sec. butyl-4-isopropyl decane

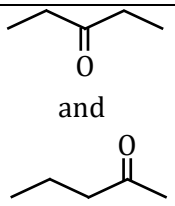
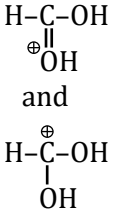
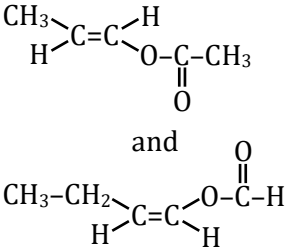
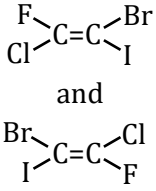
(2) 4- isopropyl-5-sec. butyl decane

(3) 6-sec. butyl-7-isopropyl decane

(4) 7-isopropyl-6-sec. butyl decane

CCN205

40. Match the column :

Column-I	Column-II
(a) 	(i) Position isomer
(b) 	(ii) Metamers
(c) 	(iii) Geometrical isomers
(d) 	(iv) Resonance structures

(1) (a) → (i); (b) → (iv); (c) → (ii); (d) → (iii)
(2) (a) → (i); (b) → (iv); (c) → (iii); (d) → (ii)
(3) (a) → (ii); (b) → (iv); (c) → (i); (d) → (iii)
(4) (a) → (iv); (b) → (i); (c) → (ii); (d) → (iii)

CIS134

41. Which is not anti aromatic ?

(1) 
(2) 
(3) 
(4) 

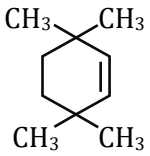
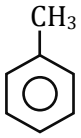
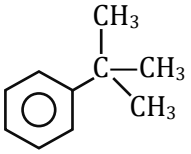
CGC280

42. **Statement-I :** Eclipsed form of n-Butane is minimum stable conformers.
Statement-II : Eclipsed form of n-Butane has maximum torsional strain but minimum steric strain.

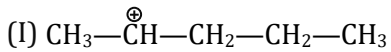
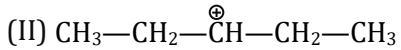
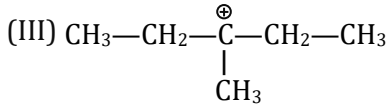
(1) Both statement I and statement II are incorrect.
(2) Both statement-I and statement-II are correct.
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct.

CIS135

43. Hyperconjugation occurs in

(1) 
(2) 
(3) 
(4) 

CGC281

44. (I) 
(II) 
(III) 

Which of the following orders is correct for the stability of these carbocations?

(1) I > II > III (2) III > II > I
(3) III > I > II (4) I > III > II

CGC282

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	T	F	T	T	F	F	T	T	T	F	T	F	F	F	F
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	T	F	F	F	T	F	T	T	T	T	4	3	3	-	-
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	
Answer	4	2	3	4	4	2	2	2	1	1	1	3	2	3	

29. (1) a > b > c
(2) a > c > b

30. Both structures have charge and less number of covalent bond than real structure.

AIM : FREE EDUCATION TO ALL APUL

Classification and Nomenclature, Isomerism & General Organic Chemistry

Direction : Q.1 To Q.25
Statement are true or false

1. Hybridization influences the bond length and bond enthalpy

CCN185

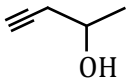
2. In $CH_3–CH=CH–CH_3$ all the atoms are in same plane because formation of π bond is possible if p orbitals on adjacent atoms are parallel.

CCN186

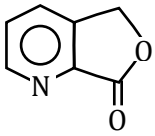
3. Number of σ bond and π bond in $CH_3–CH=CH–CN$ are 9 and 3.

CCN187

4. Bond line notation of pent-4-yn-2-ol is

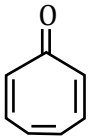


CCN188

5. Number of hetero atom in 

, are 3

CCN189

6. 

is Benzenoid structure

CCN190

7. 

(THF) is Alicyclic compound

CCN191

8. Pyridine, Furane and Thiophene all are Heterocyclic aromatic compound

CCN192

9. Ethanoic acid and Butanoic acid are Homologues and can be represented by same general molecular formula.

CCN193

10. Number of alkyl groups obtained from $(CH_3)_3C–CH(CH_3)_2$ are 4

CCN194

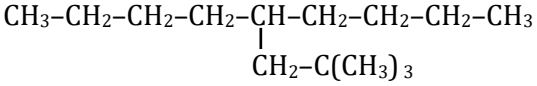
11. The correct name of 6-ethyl-3-methyl octane is 3-ethyl-6-methyl octane

CCN195

12. The name of branched chain alkyl group and their position are placed in parenthesis (in bracket) in IUPAC name

CCN196

13. The IUPAC name of

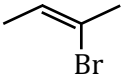


is 5-(2,2-Dimethyl ethyl) nonane

CCN197

14. If there are two different F.G. in a compound than one, which is principle functional is named as prefix and another is substituent which is used as suffix.

CCN198

15. IUPAC name of 

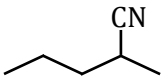
is

3-Bromobut-2-ene

CCN199

16. $CH≡C–CH=CH–CH_2–CH=CH_2$ is named as hepta-3,6-dien-1-yne

CCN200

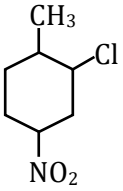
17. 

is named as 2-Cyanopentane

CCN201

18. But-2-en-3-amine is correct IUPAC name

CCN202

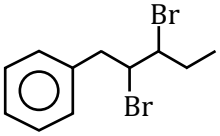
19. 

is named as

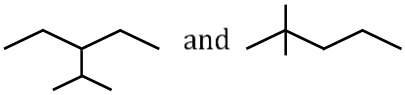
1-chloro-2-methyl-5-nitro cyclohexane

CCN203

20. The correct structure of 2,3-dibromo-1-phenyl pentane is



CCN204

21. 

are positional isomers

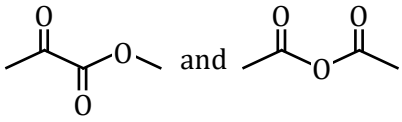
CIS131

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

22. $C_4H_{10}O$ can show metamerism

CIS132

23.  and

are functional isomers

CIS133

24. Heterolytic cleavage of $CH_3–Cu$ gives $\overset{\ominus}{C}H_3$ and $\overset{\oplus}{C}u$

CGC265

25. In $CH_3–I$ carbon is electrophilic centre

CGC266

26. The incorrect statement about Inductive effect ?

(1) In chloroethane ($CH_3–CH_2–Cl$) the polar C–Cl bond induces polarity in the adjacent bond.

(2) The effect decreases rapidity as the number of intervening bond increases.

(3) This effect becomes vanishingly small after three bonds.

(4) Aryl is electron donating groups (+I)

CGC267

27. Which is incorrect order regarding polar bond ?

(1) $CH_3–Br > CH_3–H$

(2) $CH_3–OH > CH_3–NH_2$

(3) $CH_3–CH_3 > CH_3–OH$

(4) $CH_3–F > CH_3–Cl$

CGC268

28. The incorrect statement about resonance effect?

(1) Canonical structures contribute to the actual structure in proportion to their stability.

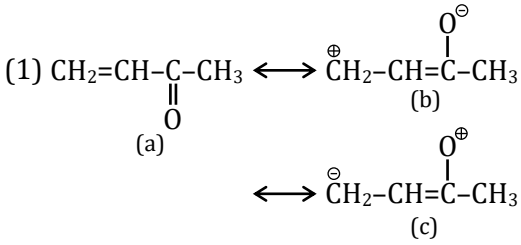
(2) The energy of actual structure is lower than that of any of the canonical structure.

(3) The difference in energy between the highest energy canonical structure and the lowest energy canonical structure is called as resonance energy.

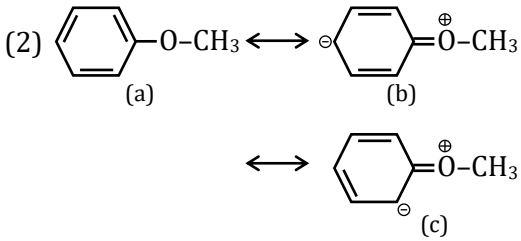
(4) The more number of canonical structure, the more is the resonance energy.

CGC269

29. Give stability order of following resonating structures

(1) 

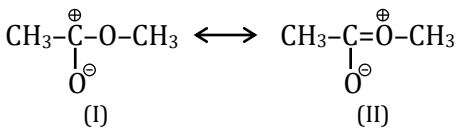
(a) (b) (c)

(2) 

(a) (b) (c)

CGC270

30. Why following two structures are not the major contributions to the real structure of $CH_3–COOCH_3$.



(I) (II)

CGC271

31. Which is ⊖R group ?

(1) $–O–\overset{\overset{O}{\|}}{C}–R$ (2) $–NH–\overset{\overset{O}{\|}}{C}–R$

(3) –Br (4) $–\overset{\overset{O}{\|}}{C}–O–R$

CGC272

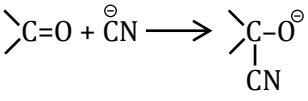
32. Correct statement about electrometric effect?

(1) Diethyl ether shows E-effect

(2) The effect is annulled as soon as the attacking reagent is removed

(3) The inductive effect predominates over E-effect

(4) The following is an example of +E-effect



CGC273

AIM : FREE EDUCATION TO ALL ♥ APUL

33. Incorrect statement about H-effect is -

(1) H-effect is permanent effect

(2) σ electron of C–H bond enter into the partial conjugation with empty p orbital.

(3) This type of overlapping destabilise the carbocation.

(4) Delocalization of electrons by hyperconjugation is also possible in alkene.

CGC274

34. $CH_3-\overset{\oplus}{C}H_2$ is more stable than $\overset{\oplus}{C}H_3$. It is explained by, except-

(1) +I of $-CH_3$ increases stability of $CH_3-\overset{\oplus}{C}H_2$

(2) Hyperconjugation interaction in $CH_3-\overset{\oplus}{C}H_2$

(3) The C–H σ bond helps in dispersing the positive charge.

(4) Electrons of C–H bond can be delocalized into empty 3p orbital.

CGC275

35. Incorrect statement about $CH_3-O-CH_2-CH_2-CH_3$ is

(1) It is unsymmetrical ether

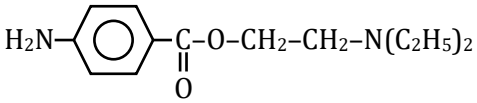
(2) It is IIIrd member of ether

(3) It is IInd member of unsymmetrical ether

(4) It is IIIrd member of unsymmetrical ether

CGC276

36. Which functional group is not present in



(1) 1° Amine (2) 2° Amine

(3) 3° Amine (4) ester

CGC277

37. Why alkyl group act as electron donor when attached to π system

(1) Due to +I effect of alkyl group

(2) Due to hyperconjugation effect

(3) (1) and (2) both

(4) Due to +M of alkyl group

CGC278

38. Which is an example of addition reaction ?

(1) $CH_3-CH_2-Br + \overset{\ominus}{H}S \rightarrow CH_3-CH_2-SH$

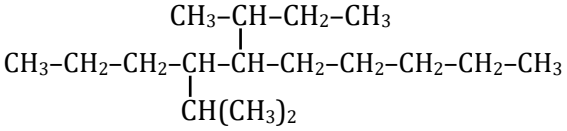
(2) $(CH_3)_2C=CH_2 + H-Cl \rightarrow (CH_3)_2\underset{\underset{Cl}{|}}{C}-CH_3$

(3) $CH_3-CH_2-Br + \overset{\ominus}{O}H \rightarrow CH_2=CH_2 + H_2O + \overset{\ominus}{Br}$

(4) $(CH_3)_3C-CH_2-OH + H-Br \rightarrow (CH_3)_2CBr\,CH_2-CH_2-CH_3 + H_2O$

CGC279

39. The correct IUPAC name of



(1) 5-sec. butyl-4-isopropyl decane

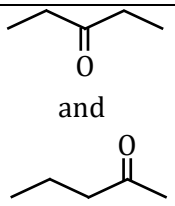
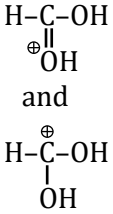
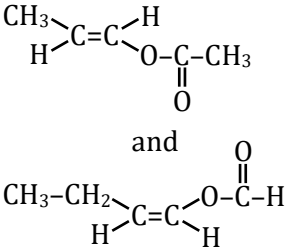
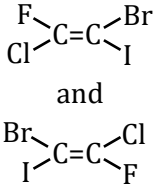
(2) 4- isopropyl-5-sec. butyl decane

(3) 6-sec. butyl-7-isopropyl decane

(4) 7-isopropyl-6-sec. butyl decane

CCN205

40. Match the column :

Column-I	Column-II
(a) 	(i) Position isomer
(b) 	(ii) Metamers
(c) 	(iii) Geometrical isomers
(d) 	(iv) Resonance structures

(1) (a) → (i); (b) → (iv); (c) → (ii); (d) → (iii)
(2) (a) → (i); (b) → (iv); (c) → (iii); (d) → (ii)
(3) (a) → (ii); (b) → (iv); (c) → (i); (d) → (iii)
(4) (a) → (iv); (b) → (i); (c) → (ii); (d) → (iii)

CIS134

41. Which is not anti aromatic ?

(1) 
(2) 
(3) 
(4) 

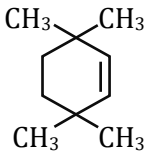
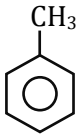
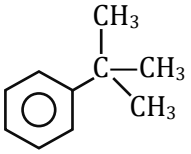
CGC280

42. **Statement-I :** Eclipsed form of n-Butane is minimum stable conformers.
Statement-II : Eclipsed form of n-Butane has maximum torsional strain but minimum steric strain.

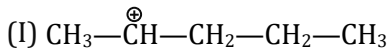
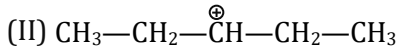
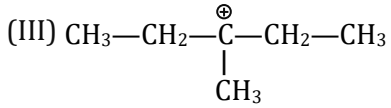
(1) Both statement I and statement II are incorrect.
(2) Both statement-I and statement-II are correct.
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct.

CIS135

43. Hyperconjugation occurs in

(1) 
(2) 
(3) 
(4) 

CGC281

44. (I) 
(II) 
(III) 

Which of the following orders is correct for the stability of these carbocations?

(1) I > II > III (2) III > II > I
(3) III > I > II (4) I > III > II

CGC282

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	T	F	T	T	F	F	T	T	T	F	T	F	F	F	F
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	T	F	F	F	T	F	T	T	T	T	4	3	3	-	-
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	
Answer	4	2	3	4	4	2	2	2	1	1	1	3	2	3	

29. (1) a > b > c
(2) a > c > b

30. Both structures have charge and less number of covalent bond than real structure.

AIM : FREE EDUCATION TO ALL APUL

Optical Isomerism

1. The sum number of chiral carbon in

(a) $CH_3-CH=CH-\underset{\underset{Cl}{|}}{CH}-CH=CH-C_2H_5$

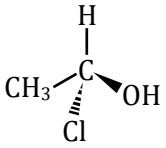
(b) $CH_3-\overset{}{\underset{}{(}}CHOH\overset{}{\underset{}{)}}_4-CH_3$

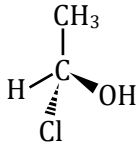
(c) $CH_3-\overset{}{\underset{}{(}}CHOH\overset{}{\underset{}{)}}_3-OH$

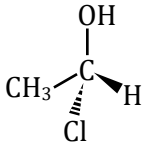
(1) 8 (2) 4 (3) 7 (4) 6

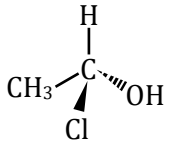
COI073

2. In the following molecule which is inversion of



(1) 

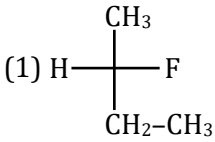
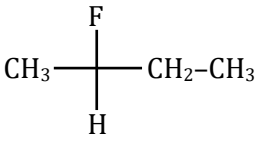
(2) 

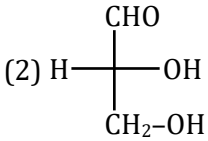
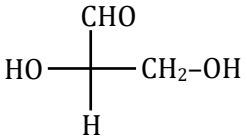
(3) 

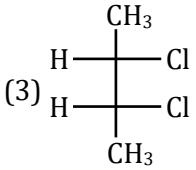
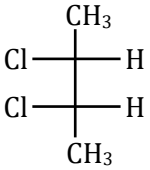
(4) All

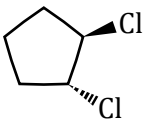
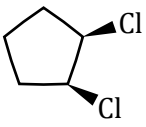
COI074

3. Which of the following pair gives racemic mixture ?

(1)  and 

(2)  and 

(3)  and 

(4)  and 

COI075

4. Incorrect statement about enantiomers is

(1) Having same physical properties like Melting point, Boiling point.

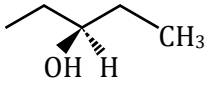
(2) They are differ in optical rotation.

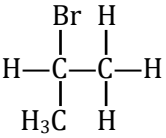
(3) If one is dextrorotatory then other will be laevorotatory.

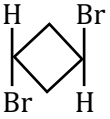
(4) Are superimposable mirror images of each other.

COI076

5. Which have no asymmetric carbon ?

(1) 

(2) 

(3) 

(4) All of them

COI077

6. Which is true statement -

(1) The sign of optical rotation is related to absolute configuration

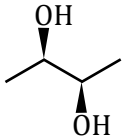
(2) Process of conversion of enantiomers into a racemic mixture is known as racemization.

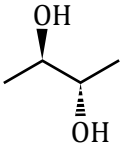
(3) Formation of a racemic mixture from non stereo isomer is known as racemization

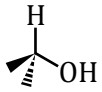
(4) Achiral compound is optically active

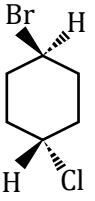
COI078

7. Which is optically active ?

(1) 

(2) 

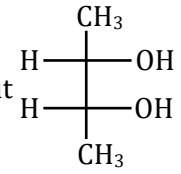
(3) 

(4) 

COI079

AIM : FREE EDUCATION TO ALL APUL

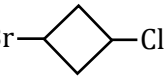
AIM : FREE EDUCATION TO ALL APUL

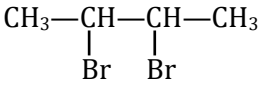
8. Incorrect statement about  is

(1) It's mirror image is superimposable on it.
(2) It is achiral molecule
(3) It's configuration is 2R, 3S
(4) It can give racemic mixture

COI080

9. Which compound can show enantiomers ?

(1) 

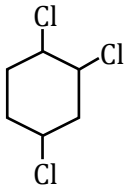
(2) 

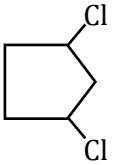
(3) 2-Methyl butane

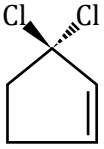
(4) All

COI081

10. Which have two chiral-C ?

(1) 2,3-Dimethylpentane (2) 

(3) 

(4) 

COI082

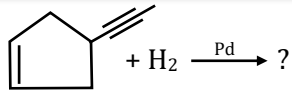
11. **Statement-I :** D and L nomenclature system is used for naming enantiomers
Statement-II : D represent dextrorotary and L represent Laevorotatory rotation of light.
(1) Both statement I and statement II are incorrect.
(2) Both statement-I and statement-II are correct.
(3) Statement-I is correct but statement-II is incorrect
(4) Statement-I is incorrect but statement-II is correct.

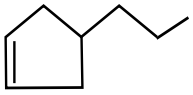
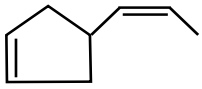
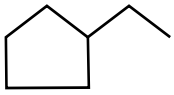
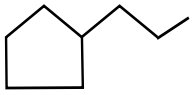
COI083

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	**3**	**4**	**1**	**4**	**4**	**2**	**1**	**4**	**2**	**3**	**3**

Hydrocarbons

1.  ?

(1) 
(2) 
(3) 
(4) 

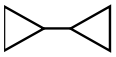
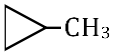
CHC321

2. $CH_3-CH_2-CH_2-Cl + H-Cl \xrightarrow{Zn}$?

(1) $CH_3-\underset{\displaystyle Cl}{\underset{|}{CH}}-CH_3$
(2) $CH_3-CH_2-CH_3$
(3) $CH_3-CH=CH_2$
(4) $CH_3-CH_2-CH\begin{matrix}Cl\\Cl\end{matrix}$

CHC322

3. $CH_3-Cl +$  $\xrightarrow[\text{dry ether}]{Na}$ products. Which is not obtained in product ?

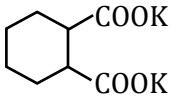
(1) CH_3-CH_3
(2) 
(3) 
(4) 

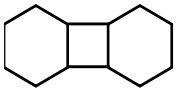
CHC323

4. Sodium salt of which acid gives propane on decarboxylation ?

(1) $CH_3-CH_2-CH_2-COONa$
(2) $CH_3-CH_2-COONa$
(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COONa$
(4) Both (1) and (3)

CHC324

5.  $\xrightarrow{\text{electrolysis}}$?

(1) 
(2) 
(3) 
(4) 

CHC325

6. Sodium salt of which acid will be needed for the preparation of butane by electrolysis?

(1) Ethanoic acid
(2) Propanoic acid
(3) Pentanoic acid
(4) Butanoic acid

CHC326

7. Arrange following in decreasing order of their boiling points.

(A) n-butane
(B) 2-methyl butane
(C) n-pentane
(D) 2,2-dimethyl propane

(1) A > B > C > D
(2) B > C > D > A
(3) C > B > D > A
(4) D > C > B > A

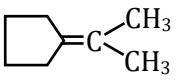
CHC327

8. Arrange following hydrogen halides in order of their decreasing reactivity with propene ?

(1) HCl > HBr > HI
(2) HBr > HI > HCl
(3) HI > HBr > HCl
(4) HCl > HI > HBr

CHC328

9. Which of the following alkene on ozonolysis give a mixture of ketones only ?

(1) $CH_3-CH=CH-CH_3$
(2) 
(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH=CH_2$
(4) $CH_3-CH=CH-CH_2-CH_2-CH_3$

CHC329

10. Arrange these halogens F_2, Cl_2, Br_2, I_2 in order of their increasing reactivity with alkanes.

(1) $I_2 < Br_2 < Cl_2 < F_2$
(2) $Br_2 < Cl_2 < F_2 < I_2$
(3) $F_2 < Cl_2 < Br_2 < I_2$
(4) $Br_2 < I_2 < Cl_2 < F_2$

CHC330

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

11. Methane reacts with steam at 1273K in the presence of Ni catalyst to form :-

(1) CO_2 (2) CO

(3) CH_3OH (4) None

CHC331

12. The increasing order of reduction of alkyl halides with zinc and dilute HCl is :-

(1) R–Cl < R –I < R–Br

(2) R–Cl < R–Br < R–I

(3) R–I < R–Br < R–Cl

(4) R–Br < R–I < R–Cl

CHC332

13. Which of the following oxidation reaction of methane is suitable example of incomplete carbocation reaction ?

(1) $CH_{4(g)} + 2O_{2(g)} \longrightarrow CO_{2(g)} + 2H_2O_{(l)}$

(2) $CH_{4(g)} + O_{2(g)} \longrightarrow C_{(s)} + 2H_2O_{(l)}$

(3) $2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn} 2CH_3COOH + 2H_2O$

(4) All of these

CHC333

14. Which reaction is suitable to obtained n pentane only in good yield ?

(1) $CH_3-CH_2-Cl + CH_3-CH_2-CH_2-Cl \xrightarrow[\text{dry ether}]{Na}$

(2) $CH_3-CH_2-CH_2-CH_2-COONa \xrightarrow[CaO/\Delta]{NaOH}$

(3) $CH_3-CH_2-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COONa \xrightarrow[CaO/\Delta]{NaOH}$

(4) $CH_3-CH_2-COONa + CH_3-CH_2-CH_2-COONa \xrightarrow{\text{electrolysis}}$

CHC334

15. From which of the following reaction methane can be obtained ?

(1) $CH_3-Cl + Na \xrightarrow[\text{ether}]{\text{dry}}$

(2) $CH_3—CH_2—COONa \xrightarrow{\text{electrolysis}}$

(3) $CH_3-Cl + Ph-Cl \xrightarrow[\text{ether}]{\text{dry}}$

(4) $CH_3-Cl \xrightarrow{Zn/HCl}$

CHC335

16. **Assertion :** Iodination of alkane is reversible.

Reason : H-I, formed in this reaction, work as an acid.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CHC336

17. Which chain propagation step gives major products in the following reaction ?

$CH_4 + Cl_2 \xrightarrow{hv}$

(1) $\dot{C}H_3 + Cl_2 \longrightarrow$ (2) $CH_3-Cl + \dot{C}l \longrightarrow$

(3) $\dot{C}H_2-Cl + Cl_2 \longrightarrow$ (4) $\dot{C}H_3 + H-Cl \longrightarrow$

CHC337

18. $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_3 + \dot{Br} \longrightarrow$

(1) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle Br}{\overset{|}{\underset{|}{C}}}}-CH_2-CH_3$ (2) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\overset{\displaystyle Br}{\overset{|}{CH}}-CH_3$

(3) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle \dot{}}{\overset{|}{\underset{|}{C}}}}-CH_2-CH_3$ (4) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\dot{C}H-CH_3$

CHC338

19. $CH_3-CH_3 + Cl_2 \xrightarrow{hv}$

The by product obtained and the step in which by product is obtained -

(1) Ethane, chain propagation

(2) Butane, chain propagation

(3) Ethyl chloride, Chain propagation

(4) Butane, Chain termination step.

CHC339

20. Which alkane is used to obtain toluene by aromatization reaction ?

(1) Iso heptane (2) 3-Methyl hexane

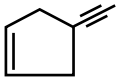
(3) 1-Chloro heptane (4) Heptane

CHC340

21. Which reaction product does not show GI ?

(1)  $+ D_2 \xrightarrow{Ni/\Delta}$

(2) $CH_3-C\equiv C-CH_3 \xrightarrow{Na/liq\ NH_3}$

(3)  $+ H_2 \xrightarrow{Ni/\Delta}$

(4) $CH_3-C\equiv C-CH_3 \xrightarrow[H_2/\Delta]{BaSO_4/Pd}$

CHC341

22. Which reaction gives two different structural product's

(1)  $+ HCl \longrightarrow$

(2) $CH_3-CH=CH-CH_3 + H-Br \xrightarrow[\Delta]{R_2O_2}$

(3) $CH_3-CH=CH_2 + H-Cl \longrightarrow$

(4) $CH_3-\underset{\underset{H_3C}{|}}{C}=\underset{\underset{CH_3}{|}}{C}-CH_3 + H-Br \longrightarrow$

CHC342

23. **Assertion :** Peroxide effect is not observed in addition of H–I.

Reason : H–I bond is weaker than H–Br and iodine free radicals combine to form I_2 instead of addition to double.

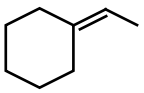
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

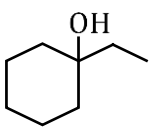
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

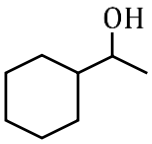
(3) Assertion is true but Reason is false.

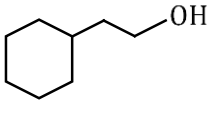
(4) Both Assertion and Reason are false.

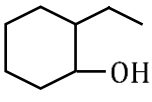
CHC343

24.  $+ H_2O \xrightarrow{H^{\oplus}}$

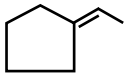
(1) 

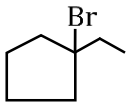
(2) 

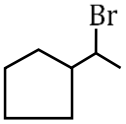
(3) 

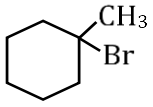
(4) 

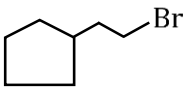
CHC344

25.  $+ H-Br \xrightarrow[\Delta]{R_2O_2}$

(1) 

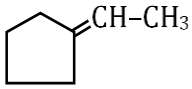
(2) 

(3) 

(4) 

CHC345

26. Which gives acetic acid on ozonolysis without zinc?

(1) 

(2) $(CH_3)_2C=CH-CH_3$

(3) $CH_3-CH=CH-CH_3$

(4) All of them

CHC346

27. In which reaction alkyne is not obtained as major product?

(1) $CH_3-\underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}}-\underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}}-CH_3 + 2Zn \longrightarrow$

(2) $CH_3-\underset{\underset{Br}{|}}{CH}-\underset{\underset{Br}{|}}{CH}-CH_3 + \text{Alc. KOH} \xrightarrow{\Delta}$

(3) $CH_3-CH_2-CH\begin{matrix}Cl\\Cl\end{matrix} + 2NaNH_2 \longrightarrow$

(4) $\underset{\underset{Br}{|}}{CH_2}-\underset{\underset{Br}{|}}{CH_2} + 2NaNH_2 \longrightarrow$

CHC347

28. **Statement–I :** 1–Alkyne and other alkyne can be differentiated by $NaNH_2$

Statement–II : 1–Alkyne doesn't react with NaOH.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CHC348

29. Addition of H–Br on 1–Butyne gives.

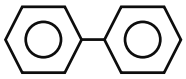
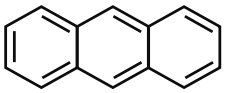
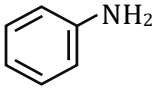
(1) Gem. dihalide (2) Vicinal dihalide

(3) Isolated dihalide (4) Mono halide

CHC349

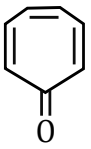
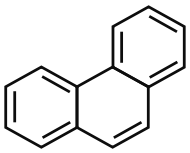
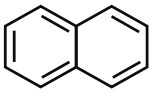
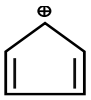
AIM : FREE EDUCATION TO ALL APUL

30. Which is an example of Non benzenoid?

(1) 
(2) 
(3) 
(4) 

CHC350

31. Which is not aromatic ?

(1) 
(2) 
(3) 
(4) 

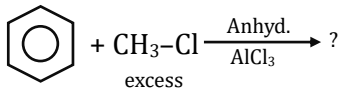
CHC351

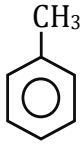
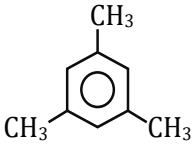
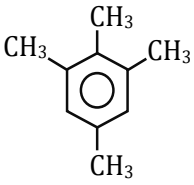
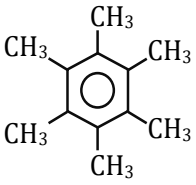
32. **Assertion :** Reaction of 1–Chloropropane and benzene gives isopropyl benzene not n propyl benzene

Reason : Reaction occurs via rearrangement in carbocation.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CHC352

33. 

(1) 
(2) 
(3) 
(4) 

CHC353

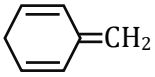
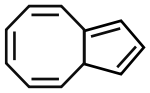
34. **Statement–I :** $-NO_2$ on Benzene is meta directing group.

Statement–II : The electron density on –O and –P is decreased and at meta is increased.

(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

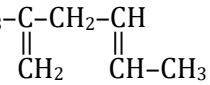
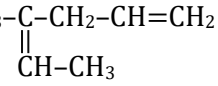
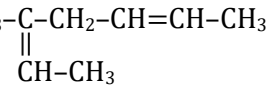
CHC354

35. Which is aromatic ?

(1) 
(2) 
(3) 
(4) 

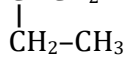
CHC355

36. The structure of reactant which gives H–CHO, CH_3–CHO and CH_3–$COCH_2$–CHO on ozonolysis ?

(1) 
(2) 
(3) 
(4) Both (1) and (2)

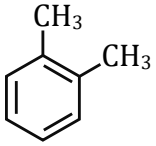
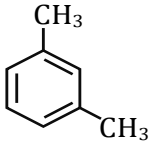
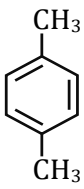
CHC356

37. Which gives 1 mole of CO_2 and 2 moles of CH_3–COOH on oxidation with acidic $KMnO_4$?

(1) $CH_3-CH=CH-CH_3$
(2) 
(3) $CH_3-CH=C=CH-CH_3$
(4) 

CHC357

38. Which gives one moles of CHO–CHO, and two moles of CH_3–CO–CHO on ozonolysis ?

(1) 
(2) 
(3) 
(4) All of them

CHC358

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

39. The structure of alkene which on hydrogenation give 2–methyl pentane ?

(1) $CH_2=\underset{\displaystyle CH_3}{\overset{|}{C}}-CH_2-CH_2-CH_3$

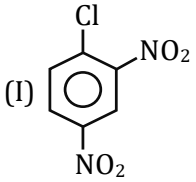
(2) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}=CH-CH_2-CH_3$

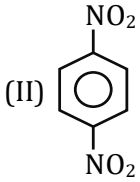
(3) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{CH}}-CH=CH_2-CH_3$

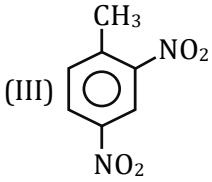
(4) All of them

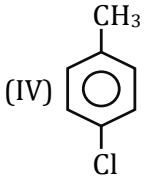
CHC359

40. Correct reactivity order towards an electrophile ?

(I) 

(II) 

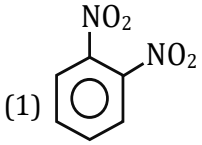
(III) 

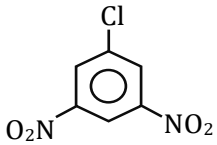
(IV) 

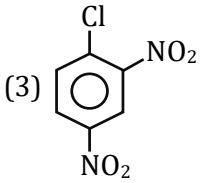
(1) (IV) > (I) > (II) > (III)
(2) (IV) > (I) > (III) > (II)
(3) (IV) > (III) > (II) > (I)
(4) (III) > (IV) > (I) > (II)

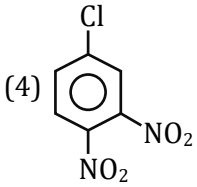
CHC360

41. Dinitration of chlorobenzene gives ?

(1) 

(2) 

(3) 

(4) 

CHC361

42. Given below are two statements, one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Benzene is more stable than hypothetical cyclohexatriene.

Reason R : The delocalized π electron cloud is attracted more strongly by nuclei of carbon atoms.

In the light of the above statements, choose the correct answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are correct and R is the correct explanation of A.

(4) Both A and R are correct but R is not the correct explanation of A.

CHC362

43. Given below are two statements:

Statement-I : Peroxide effect is not observed in addition of HCl & HI

Statement-II : HCl bond is stronger than HBr bond and H – I bond is weaker than H – Br bond.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

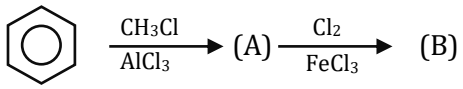
(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

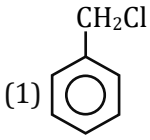
(4) Statement-I and II both are incorrect.

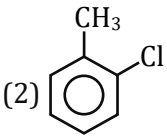
CHC363

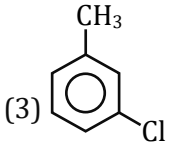
44. For the reaction

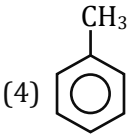


Product B is :

(1) 

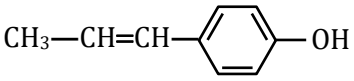
(2) 

(3) 

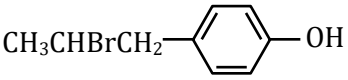
(4) 

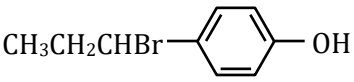
CHC364

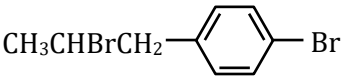
45. The reaction of :

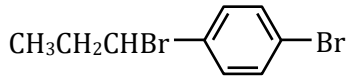


with HBr gives :

(1) 

(2) 

(3) 

(4) 

CHC365

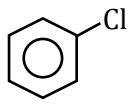
ANSWER KEY

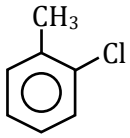
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**4**	**4**	**3**	**2**	**3**	**3**	**2**	**1**	**2**	**2**	**2**	**3**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**1**	**3**	**4**	**4**	**3**	**3**	**1**	**1**	**2**	**4**	**2**	**1**	**1**	**2**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**4**	**1**	**4**	**3**	**3**	**4**	**3**	**1**	**4**	**3**	**3**	**3**	**1**	**2**	**2**

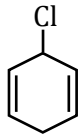
AIM : FREE EDUCATION TO ALL APUL

Haloalkanes and Haloarenes

1. Which is not an aryl chloride?

(1) 

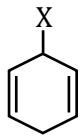
(2) 

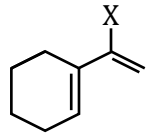
(3) 

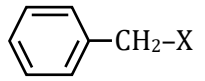
(4) All

CHH244

2. Which is true statement ?

(1)  is allylic halide

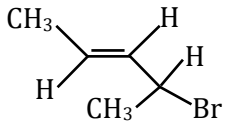
(2)  is vinylic halide

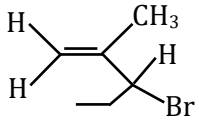
(3)  is Benzylic Halide

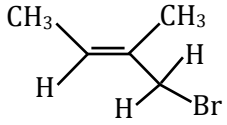
(4) All

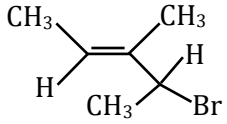
CHH245

3. Which is 4-Bromo-3-Methylpent-2-ene ?

(1) 

(2) 

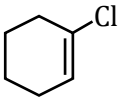
(3) 

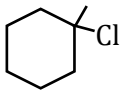
(4) 

CHH246

4. Which is/are 2° halide

(a) $CH_2=CH-\underset{\displaystyle CH_3}{\underset{|}{CH}}-Cl$ (b) $CH_2=CH-Cl$

(c) $Ph-\underset{\displaystyle Cl}{\underset{|}{CH}}-Ph$ (d) 

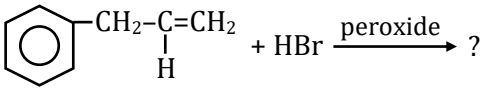
(e) 

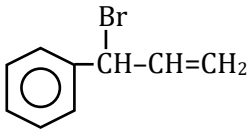
Correct option is

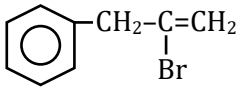
(1) a, c, and d (2) a, c, d and e

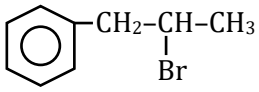
(3) b and d (4) a and d

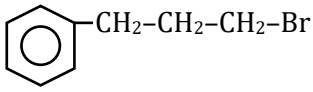
CHH247

5. 

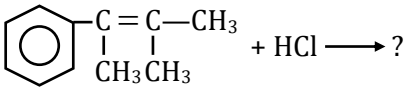
(1) 

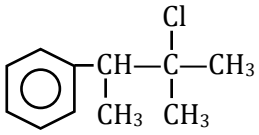
(2) 

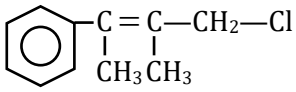
(3) 

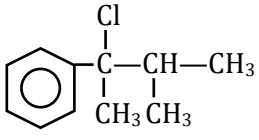
(4) 

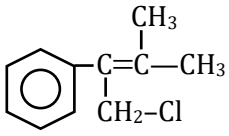
CHH248

6. 

(1) 

(2) 

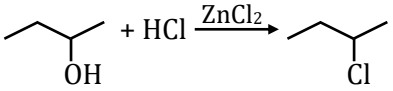
(3) 

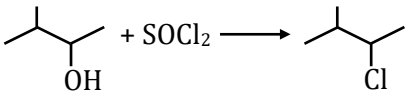
(4) 

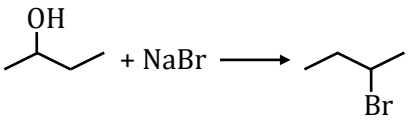
CHH249

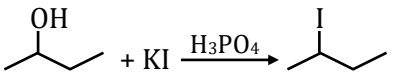
AIM : FREE EDUCATION TO ALL APUL

7. Which of the following is not correct method to obtain halide

(1) 

(2) 

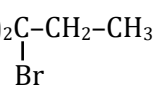
(3) 

(4) 

CHH250

8. Incorrect statement about the following reaction ?

$(CH_3)_2CH-CH_2-CH_3 + Br_2 \xrightarrow{hv}$

(1) Major product is 

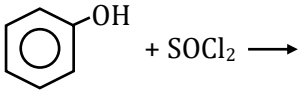
(2) Total structural monobromo derivative obtained are 4

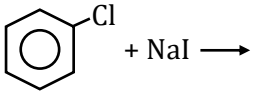
(3) Reaction is free radical substitution

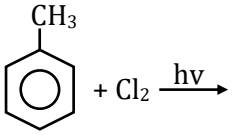
(4) None

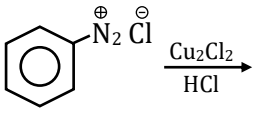
CHH251

9. Which reaction gives aryl halide as a major product ?

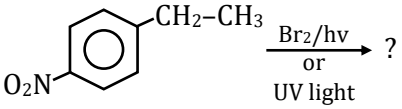
(1) 

(2) 

(3) 

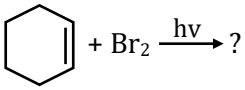
(4) 

CHH252

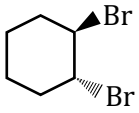
10. 

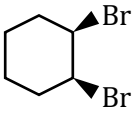
Major product is

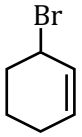
CHH253

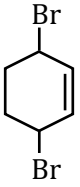
11. 

Major product is ?

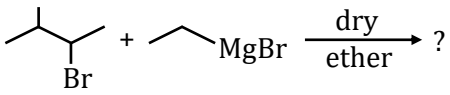
(1) 

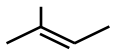
(2) 

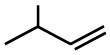
(3) 

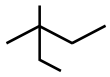
(4) 

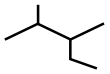
CHH254

12. 

(1) 

(2) 

(3) 

(4) 

CHH255

13. Which reaction does not give cyanide

(1) $CH_3-CH_2-Cl + KCN \longrightarrow$

(2) $Ph-CH_2-OH + NaCN \longrightarrow$

(3) $Ph-CH_2-Cl + AgCN \longrightarrow$

(4) Both (2) and (3)

CHH256

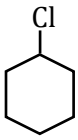
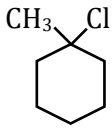
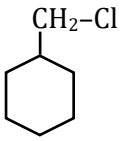
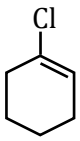
AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

14. Reaction of Haloalkane with AgCN gives isocyanide as the major product not cyanide due to -
(1) C–N bond is more stable than C–C bond
(2) AgCN is mainly covalent and carbon is free to donate electron pair
(3) AgCN is mainly ionic and carbon is free to donate electron pair
(4) AgCN is mainly covalent bond and nitrogen is free to donate electron pair

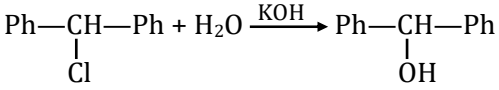
CHH257

15. Most reactive towards S_N2 is

(1) 
(2) 
(3) 
(4) 

CHH258

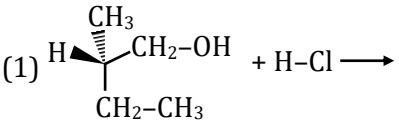
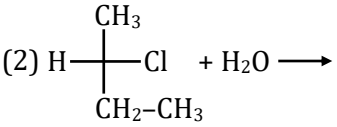
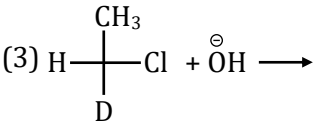
16. Incorrect statement about following reaction is



(1) It occurs in two steps
(2) Reaction is Biomolecule substitution
(3) Rate of reaction depends on concentration of only one reactant that is halide.
(4) Reaction intermediate is carbocation.

CHH259

17. Which is an example of pure inversion

(1) 
(2) 
(3) 
(4) All

CHH260

18. Which is the example of racemization process?
(1) $Ph-CH_2-Cl + H_2O \longrightarrow$
(2) $Ph-\underset{\underset{Cl}{|}}{CH}-CH_3 + H_2O \longrightarrow$
(3) $CH_3-CH_2-CH=CH_2 + HCl \longrightarrow$
(4) $CH_3-\underset{\underset{Cl}{|}}{CH}-D + \overset{\ominus}{O}H \longrightarrow$

CHH261

19. $CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{Br}{|}}{CH}-CH_3 + (CH_3)_3C-\overset{\ominus\oplus}{O}K \longrightarrow ?$

(1) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH-CH_3$
(2) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2$
(3) $CH_3-\underset{\underset{H_3C}{|}}{CH}-\underset{\underset{OC(CH_3)_3}{|}}{CH}-CH_2$
(4) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{OC(CH_3)_3}{|}}{C}}-CH_2-CH_3$

CHH262

20. Which can give only one alkene when heated with KOH ?
(1) 2-Bromopentane
(2) 2-Bromo-2-methyl propane
(3) 2-Bromo-3-methyl butane
(4) Neopentyl bromide

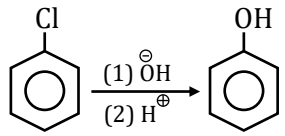
CHH263

21. Correct statement regarding Grignard reagent
(1) Carbon – magnesium bond is highly polar covalent bond
(2) Carbon – magnesium bond is ionic bond
(3) It reacts with Na in aq. medium to give alkane as a major product.
(4) It reacts with ethyl chloride to give ethene as a major product.

CHH264

22. Aryl halide is less reactive than aliphatic halide towards NSR due to -
(1) The electron pairs on halogen are in conjugation with π bond of ring.
(2) Electron rich nucleophile does not attack easily on electron rich arenes.
(3) The sp^2C of ring holds the electron pair of C–X bond more tightly.
(4) All

CHH265

23. 

Incorrect statement about above reaction is

(1) Reaction is nucleophilic substitution.

(2) Presence of electron withdrawing group increases reactivity of haloarenes.

(3) The effect is pronounced when $-NO_2$ is introduced at meta position.

(4) Intermediate anion is formed

CHH266

24. **Statement-I :** In nucleophilic substitution of haloarenes $-NO_2$ at ortho and para increases stability of intermediate anion, compared to $-NO_2$ at meta position.

Statement-II : –M of $-NO_2$ is observed at ortho and para position not at meta position.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CHH267

25. **Assertion :** Haloarenes gives NSR at high temperature but 2,4,6 trinitro haloarene gives NSR at low warm.

Reason : A nucleophile can attack easily when–M group is attached to ring.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CHH268

26. **Assertion :** Cl is deactivating group but if Haloarene is nitrated then major product is para nitro haloarene not meta.

Reason : Reactivity is controlled by the stronger inductive effect of halogen (–I > +M) and direction influence is controlled by resonance effect.

(1) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CHH269

27. Incorrect statement is/are is

(1) Grignard reagent should be prepared under anhydrous conditions

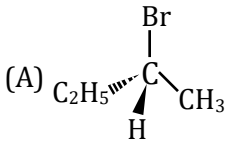
(2) n Butyl bromide reacted with KCN gives sec. butyl cyanide

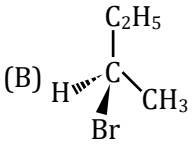
(3) Out of $C_6H_5-CHCl-C_6H_5$ and $C_6H_5-CH_2-Cl$, $C_6H_5-CHCl-C_6H_5$ is easily hydrolysed.

(4) Alkyl halides are polar but immisible with water.

CHH270

28. The addition of HBr to 1-butene gives mixture of products A, B and C.

(A) 

(B) 

(C) $CH_3-CH_2-CH_2-CH_2-Br$

The mixture consists of :-

(1) B as major, A and C as minor products

(2) A and B as major and C as major products

(3) B as major, A and C as major products

(4) A and B as major and C as minor products

CHH271

29. **Assertion :** $Ph-CH_2-Cl$ gives racemic mixture on hydrolysis.

Reason : Reaction follows via S_N2.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

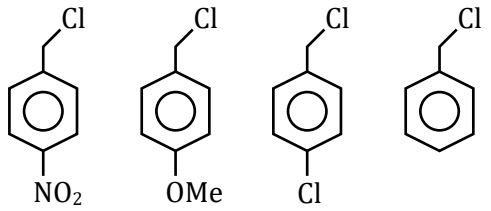
(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CHH272

AIM : FREE EDUCATION TO ALL APUL

30. Decreasing order towards S_N1 reaction for the following compounds is:



(a) (b) (c) (d)

(1) a > c > d > b (2) a > b > c > d

(3) b > d > c > a (4) d > b > c > a

CHH273

31. In S_N2 reactions on asymmetric centres

(1) inversion of configuration occurs because nucleophile attacks from the back-side of leaving group

(2) retention of configuration occurs because nucleophile attacks from the front side of leaving group

(3) both retention and inversion occur in equal proportions. It means a racemic mixture is produced

(4) the chiral centre becomes achiral

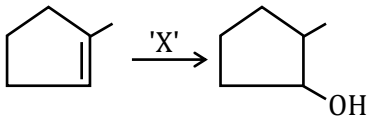
CHH274

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**4**	**4**	**1**	**4**	**3**	**3**	**4**	**4**	**3**	**3**	**4**	**4**	**4**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**2**	**3**	**2**	**2**	**2**	**1**	**4**	**3**	**1**	**1**	**1**	**2**	**4**	**4**	**3**
Question	31														
Answer	**1**														

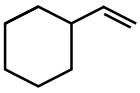
AIM : FREE EDUCATION TO ALL APUL

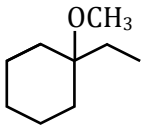
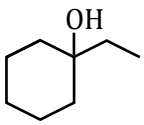
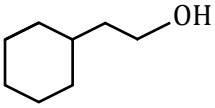
Alcohols, Phenols and Ethers

1.  Reagent 'X' is

(A) (I) BH_3/THF ; (II) $H_2O_2/\overset{\ominus}{O}H$
(B) (I) $H-Br/R_2O_2$; (II) H_2O
(C) (I) $H-Cl/R_2O_2$; (II) H_2O
(D) (I) $H-Br/R_2O_2$; (II) $\overset{\ominus}{O}H$
Correct choice is
(1) A, B and D (2) A and D only
(3) B and D only (4) All

CAE189

2. A compound  is converted in compound listed in Column-I. The suitable reagent is listed in column-II.

Column-I (Product)	Column-II (Reagent)
(A) 	(P) $H^{\oplus}/H_2O$
(B) 	(Q) (i) $H-Cl/R_2O_2$ (ii) H_2O
(C) 	(R) (i) BH_3/THF (ii) $H_2O_2/\overset{\ominus}{O}H$
	(S) (i) $H^{\oplus}$ (ii) CH_3OH

Correct Match is
(1) A → S ; B → P, Q ; C → R
(2) A → Q ; B → P ; C → R, S
(3) A → S ; B → P ; C → Q, R
(4) A → Q ; B → P ; C → R

CAE190

3. In which reaction 1° Alcohol is not obtained ?

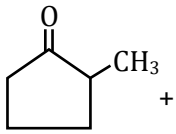
(1) $CH_3-COOH \xrightarrow[(2)H_2O]{(1)LiAlH_4}$

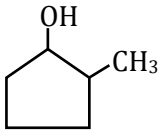
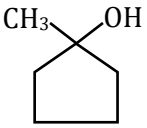
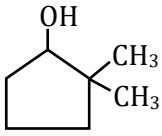
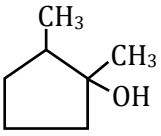
(2) $CH_3-COOCH_3 \xrightarrow[Ni]{H_2/\Delta}$

(3) $CH_3-CHO + CH_3MgBr \xrightarrow{H_2O}$

(4) $H-CHO + EtMgBr \xrightarrow{H_2O}$

CAE191

4.  $+ CH_3MgBr \xrightarrow{H_2O}$?

(1) 
(2) 
(3) 
(4) 

CAE192

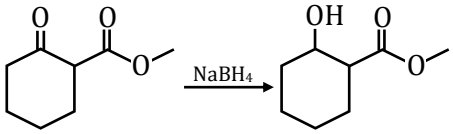
5. Which of the following reaction gives $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-Ph$ as a product.

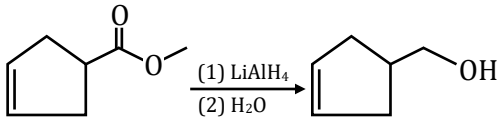
(1) $CH_3-CHO + Ph-CH_2MgBr \xrightarrow{H_2O}$
(2) $Ph-CH_2-CHO + CH_3MgBr \xrightarrow{H_2O}$
(3) $Ph-CHO + CH_3-CH_2MgBr \xrightarrow{H_2O}$
(4) (1) and (2) both

CAE193

6. Which is incorrect reaction regarding major product ?

(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CHO \xrightarrow{NaBH_4}$ 2° Alcohol

(2) 

(3) 

(4) $CH_3-\underset{\underset{CH_3}{|}}{CH}MgBr + H-CHO \xrightarrow{H_2O} CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-OH$

CAE194

7. Which can be most easily protonated
(1) CH_3-CH_2-OH (2) CH_3-O-CH_3
(3) $Ph-O-CH_3$ (4) $CH_3-CH_2-O-CH_3$

CAE195

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

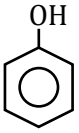
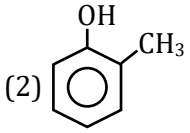
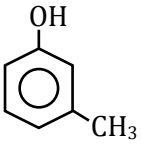
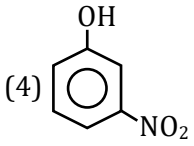
8. Which is incorrect statement ?
(1) Protonated alcohol act as electrophile
(2) Alcohol is neutral in nature
(3) Alcohol is nucleophile
(4) Alcohol and phenol both can react with NaOH

CAE196

9. **Assertion :** Alkoxide ion is better proton acceptor than hydroxide ion
Reason : Water is a better proton donor than alcohol.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CAE197

10. Which is less acidic than para cresol ($p-CH_3-C_6H_4-OH$)

(1) 
(2) 
(3) 
(4) 

CAE198

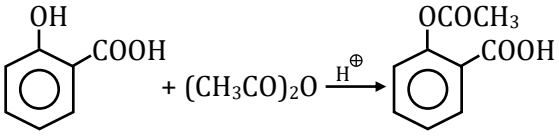
11. **Statement-I :** Esterification reaction of alcohol with carboxylic acid is carried in presence of H_2SO_4.
Statement-II : The reaction is reversible.
(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

CAE199

12. **Assertion :** Reaction of acetyl chloride with alcohol is carried in presence of base (pyridine).
Reason : It shifts the equilibrium to the right hand side by removing HCl.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CAE200

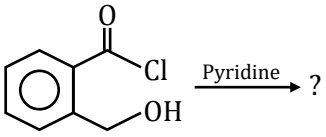
13. The incorrect statement regrading following reaction ?

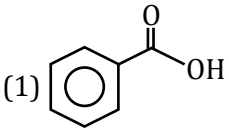
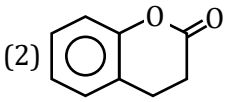
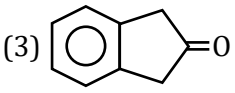
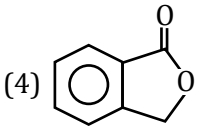


(Aspirin)

(1) Product is used as analgesic
(2) Types of reaction is nucleophilic substitution.
(3) There is acetylation of salicylic acid
(4) Acidic medium decreases the rate of reaction

CAE201

14. 

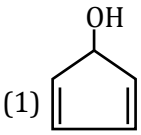
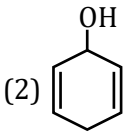
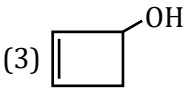
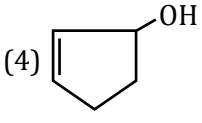
(1) 
(2) 
(3) 
(4) 

CAE202

15. **Statement-I :** Tertiary alcohol produce turbidity immediately with Lucas reagent
Statement-II : Primary alcohols do not produce turbidity at room temperature .
(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

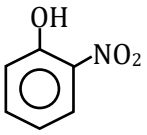
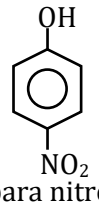
CAE203

16. Which is most reactive towards acid catalysed dehydration ?

(1) 
(2) 
(3) 
(4) 

CAE204

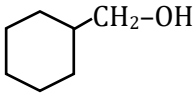
17. Incorrect statement about two statements

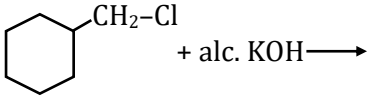
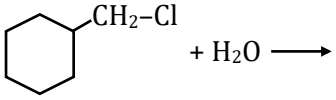
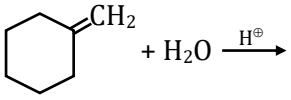
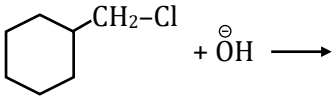



Ortho nitro phenol **(A)** and para nitro phenol **(B)**

(1) B is more acidic than A
(2) B is steam volatile
(3) Mixture of A and B can be separated by steam distillation
(4) Boiling point order → B > A

CAE205

18. In Which reaction  is obtained as major product ?

(1) 
(2) 
(3) 
(4) 

CAE206

19. Match the column-I with Column-II

Column-I	Column-II (Reagent)
(A) CH_3-CH_2-OH → Carboxylic acid	(P) Cu/Δ
(B) $CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-CH_2-OH$ → Aldehyde	(Q) $KMnO_4/H^{\oplus}$
(C) $CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{\overset{\displaystyle CH_3}{\overset{\vert}{C}}}}-OH$ → Alkene	(R) PCC

(1) A → R; B → Q; C → P
(2) A → P; B → Q; C → R
(3) A → R; B → P; C → Q
(4) A → Q; B → R; C → P

CAE207

20. **Assertion :** Preparation of ethers by acid dehydration of secondary and tertiary alcohols is not a suitable method.
Reason : Secondary and tertiary alcohols follows S_N1 pathway.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CAE208

21. In which reaction ether is obtained as a major product

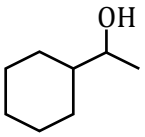
(1) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{\vert}{\underset{\vert}{C}}}}-CH_2-Cl + CH_3-\overset{\ominus}{O}-\overset{\oplus}{Na} \longrightarrow$

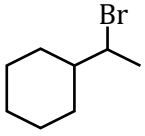
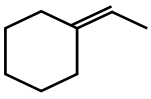
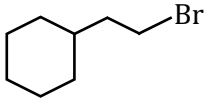
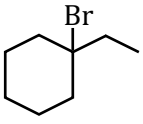
(2) $CH_3-Cl + CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{\vert}{\underset{\vert}{C}}}}-\overset{\ominus}{O}-\overset{\oplus}{Na} \longrightarrow$

(3) $Ph-Cl + CH_3-CH_2-\overset{\ominus}{O}-\overset{\oplus}{Na} \longrightarrow$

(4) $CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-Cl + CH_3-CH_2-\overset{\ominus}{O}-\overset{\oplus}{Na} \longrightarrow$

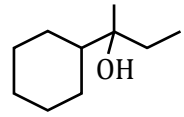
CAE209

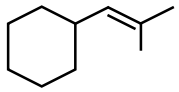
22.  + H–Br ⟶ ?

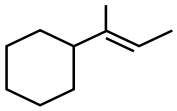
(1) 
(2) 
(3) 
(4) 

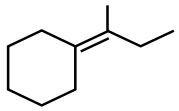
CAE210

AIM : FREE EDUCATION TO ALL APUL

23.  can be obtained by reaction of dil. H_2SO_4

(1) 

(2) 

(3) 

(4) (2) and (3) both

CAE211

24. $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow{Na} A \xrightarrow{C_2H_5Br} B$?

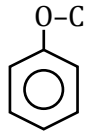
(1) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{CH_3}{|}}{CH}-OC_2H_5$

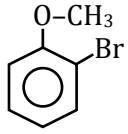
(2) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2$

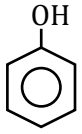
(3) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{C}=CH-CH_3$

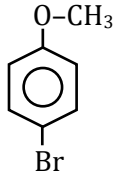
(4) $CH_3-CH_2-\underset{\underset{CH_2}{||}}{C}-CH_2-CH_3$

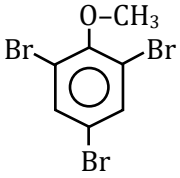
CAE212

25.  $+ Br_2 \xrightarrow[CH_3-COOH]{Fe}$ Major product

(1) 

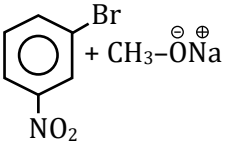
(2) 

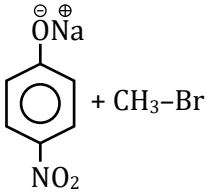
(3) 

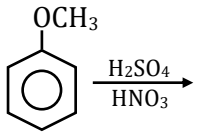
(4) 

CAE213

26. Which is appropriate set of reactants for the preparation of 1-Methoxy-4-nitrobenzene

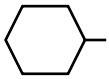
(1) 

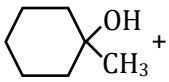
(2) 

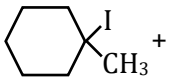
(3) 

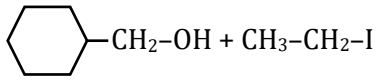
(4) (2) and (3) Both

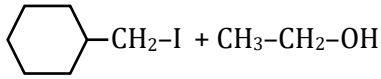
CAE214

27.  $+ H-I \longrightarrow$?

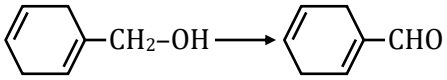
(1) 

(2) 

(3) 

(4) 

CAE215

28. 

Suitable reagent for above conversation is

(1) $H^{\oplus}/KMnO_4$ (2) $H^{\oplus}/K_2Cr_2O_7$

(3) CrO_3 (4) PCC

CAE216

29. $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}}-CH_2-CH_3 \xrightarrow[(ii)\ KMnO_4]{(i)\ H^{\oplus}/\Delta}$?

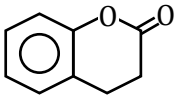
(1) $CH_3-\underset{\underset{CH_3}{|}}{C}=CH-CH_3$

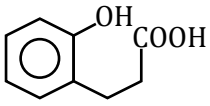
(2) $CH_3-COOH + H-COOH$

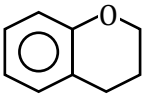
(3) Only CH_3-COOH

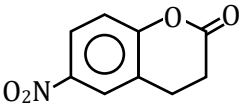
(4) $CH_3-\underset{\underset{O}{||}}{C}-CH_3 + CH_3-COOH$

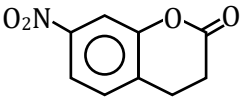
CAE217

30.  $\xrightarrow[HNO_3]{H_2SO_4}$?

(1) 

(2) 

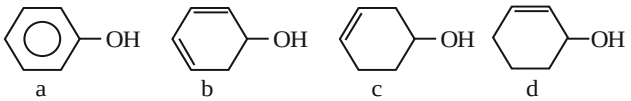
(3) 

(4) 

CAE218

AIM : FREE EDUCATION TO ALL APUL

31. Decreasing order of dehydration of the following alcohols is



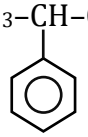
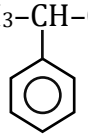
(1) a > d > b > c (2) b > d > c > a
(3) b > a > d > c (4) d > b > c > a

CAE219

32. Sulphuric acid is not used during the reaction of alcohol with KI because ?

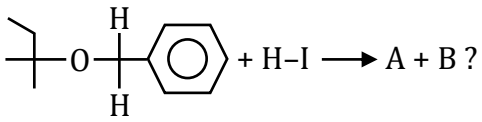
(1) $I^{\ominus}$ is strong reducing agent which reduce H_2SO_4 and converts in I_2.
(2) $I^{\ominus}$ is strong reducing agent which reduce H_2SO_4 and converts in H-I.
(3) Alcohol gets dehydrated.
(4) Oxidation of alcohol is possible.

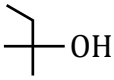
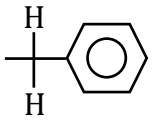
CAE220

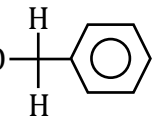
33. $CH_3-CH-CH=CH_2$  $\xrightarrow{A}$ $CH_3-CH-CH_2-CH_2-OH$ 

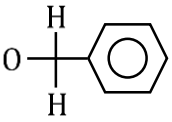
A will be
(1) H_2O/H^+
(2) $BH_3.THF/H_2O_2/\overset{\ominus}{O}H$
(3) Both
(4) None of these

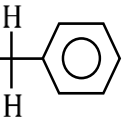
CAE221

34. 

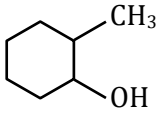
(1) A =  ; B = 

(2) A =  ; B = 

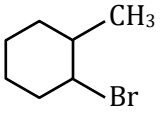
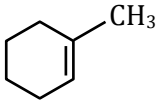
(3) A =  ; B = 

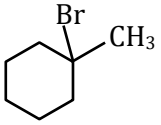
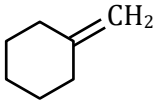
(4) A =  ; B = 

CAE222

35.  $\xrightarrow{HBr}$ P + H_2O (Major)

P should be

(1)  (2) 

(3)  (4) 

CAE223

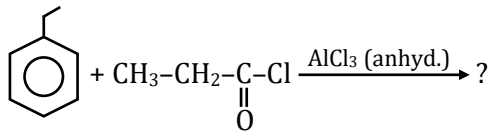
ANSWER KEY

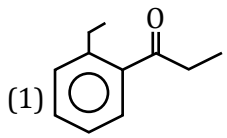
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**1**	**3**	**4**	**4**	**1**	**4**	**4**	**1**	**2**	**1**	**1**	**4**	**4**	**1**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**2**	**2**	**4**	**4**	**2**	**2**	**4**	**4**	**1**	**3**	**4**	**3**	**4**	**4**	**3**
Question	31	32	33	34	35										
Answer	**2**	**1**	**2**	**1**	**3**										

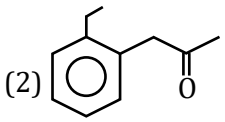
AIM : FREE EDUCATION TO ALL APUL

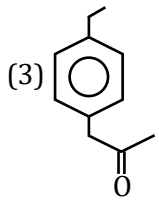
Aldehydes, Ketones and Carboxylic acids

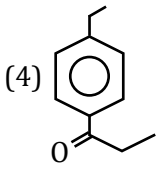
AIM : FREE EDUCATION TO ALL APUL

1. 

(1) 

(2) 

(3) 

(4) 

CAA286

2. $2CH_3-CH_2-\underset{\underset{O}{\|}}{C}-Cl + (C_6H_5-CH_2)_2Cd \longrightarrow ?$

(1) $CH_3-\underset{\underset{CH_2-Ph}{|}}{CH}-COCl$

(2) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-CH_2-Ph$

(3) $CH_3-CH_2-\underset{\underset{OH}{|}}{CH}-CH_2-Ph$

(4) $CH_3-CH_2-\underset{\underset{OH}{|}}{\overset{\overset{CH_2-Ph}{|}}{C}}-CH_2-Ph$

CAA287

3. Correct order of boiling point of

(I) $CH_3-CH_2-CH_2-CHO$

(II) $CH_3-CH_2-COCH_3$

(III) $CH_3-CH_2-CH_2-CH_2-OH$

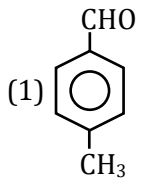
(IV) $C_2H_5-O-C_2H_5$

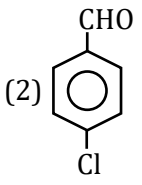
(1) III > I > II > IV (2) III > II > I > IV

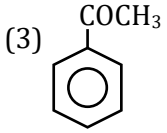
(3) II > I > III > IV (4) I > II < IV > III

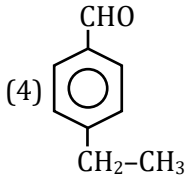
CAA288

4. Which have maximum electrophilicity ?

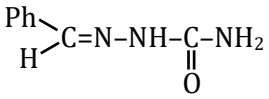
(1) 

(2) 

(3) 

(4) 

CAA289

5. $Ph-CHO + Reagent \xrightarrow{H^{\oplus}}$ 

(1) $NH_2-NH-\underset{\underset{O}{\|}}{C}-NH-NH_2$

(2) $NH_2-\underset{\underset{O}{\|}}{C}-NH-NH_2$

(3) $NH_2-NH-\underset{\underset{O}{\|}}{C}-NH-Cl$

(4) $NH_2-\underset{\underset{O}{\|}}{C}-NH-NHCH_3$

CAA290

6. **Statement-I :** Reaction of Aldehyde/Ketone with ammonia derivative is catalysed by acid and is reversible.

Statement-II : Reaction equilibrium favours the product formation if H_2O is removed rapidly.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CAA291

7. Match the column-I with column-II

Column-I	Column-II
(A) $>C=O$ + NH_3	(P) Oxime
(B) $>C=O$ + $R-NH_2$	(Q) Imine
(C) $>C=O$ + H_2N-OH	(R) Phenyl hydrazone
(D) $>C=O$ + $NH_2-NH-Ph$	(S) Schiff base

(1) A → S; B → Q; C → P; D → R

(2) A → Q; B → S; C → P; D → R

(3) A → S; B → Q; C → R; D → P

(4) A → Q; B → S; C → R; D → P

CAA292

AIM : FREE EDUCATION TO ALL APUL

8. Ethyl methyl ketone on oxidation by strong oxidising agent at elevated temperature gives?

(1) Ethanoic acid (2) Propanoic acid

(3) Methanoic acid (4) Butanoic acid

CAA293

9. **Statement-I :** Aromatic aldehyde do not give fehling's test.

Statement-II : Mild oxidising agent like Tollen's reagent, fehling solution are used to distinguish aldehyde and ketone.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CAA294

10. **Assertion :** Among all aldehyde (Aliphatic or aromatic) only acetaldehyde gives iodoform test.

Reason : Other aldehyde can not be oxidised by NaOI.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CAA295

11. Correct statement about iodoform test is

(a) All methyl ketone gives iodoform test

(b) $CH_3-\underset{\underset{OH}{|}}{CH}-$ type of alcohol gives iodoform test

(c) The oxidation does not affect a carbon-carbon double bond.

(d) $CH_3-\overset{\overset{O}{\|}}{C}-CH_2-\overset{\overset{O}{\|}}{C}-O-CH_3$ can give iodoform test.

(1) All are correct (2) Only a and c

(3) a, b, c (4) a, c, d

CAA296

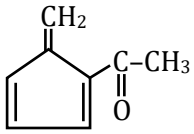
12. An organic compound with M.F. C_8H_8O gives following properties

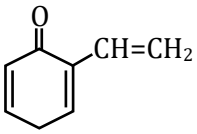
(a) Orange red ppt with 2,4-DNP

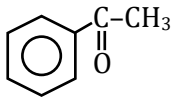
(b) Yellow ppt on heating with I_2/NaOH

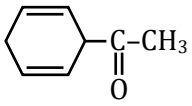
(c) Not gives Tollen's test and bromine water test.

The compound C_8H_8O

(1) 

(2) 

(3) 

(4) 

CAA297

13. **Statement-I :** α-hydrogen of aldehyde and ketone can be removed by strong base.

Statement-II : The acidity of α-hydrogen is due to strong electron withdrawing effect of carbonyl group and resonance stabilization of conjugate base

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CAA298

14. **Statement-I :** An aldehyde having at least one α-hydrogen, gives α-hydroxy aldehyde (Aldol) in presence of dil. NaOH.

Statement-II : The aldol readily lose water to give α, β-unsaturated carbonyl compound.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CAA299

15. Which is not cross-aldol product of

$CH_3-CH_2-CHO + Ph-CH_2-CHO \xrightarrow[2.\ \Delta]{1.\ NaOH}$

(1) $CH_3-CH_2-CH=\underset{\underset{CH_3}{|}}{C}-CHO$

(2) $CH_3-CH=\underset{\underset{CH_3}{|}}{C}-CHO$

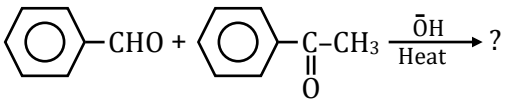
(3) $CH_3-CH_2-CH=CH-CHO$

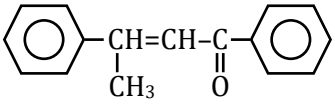
(4) All of them

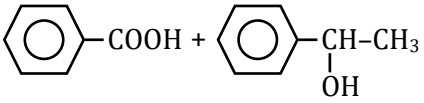
CAA300

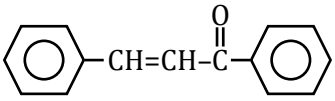
AIM : FREE EDUCATION TO ALL APUL

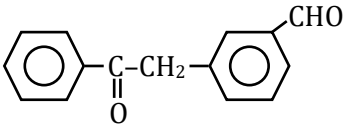
16. Major product of



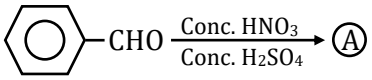
(1) 

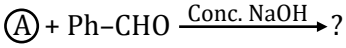
(2) 

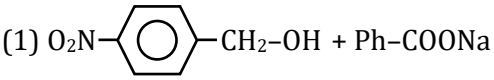
(3) 

(4) 

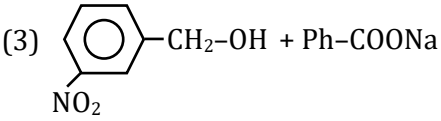
CAA301

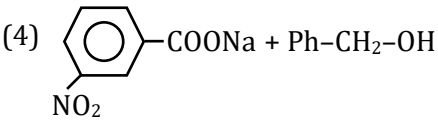
17. 



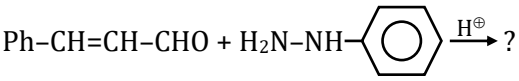
(1) 

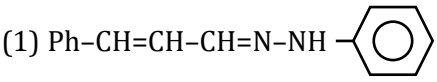
(2) 

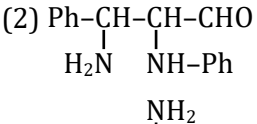
(3) 

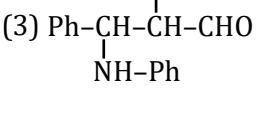
(4) 

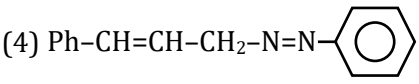
CAA302

18. 

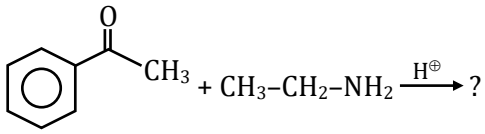
(1) 

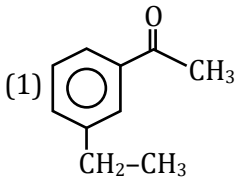
(2) 

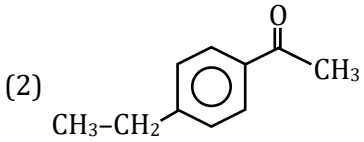
(3) 

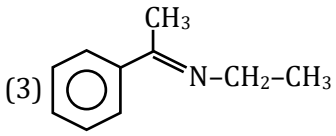
(4) 

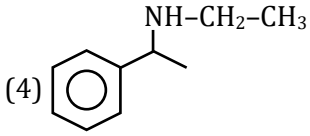
CAA303

19. 

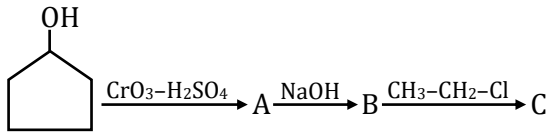
(1) 

(2) 

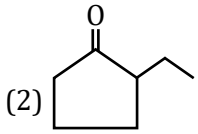
(3) 

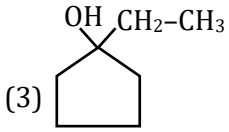
(4) 

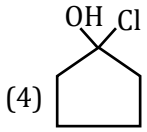
CAA304

20. 

(1) 

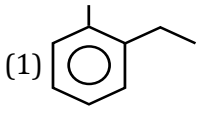
(2) 

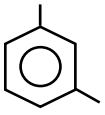
(3) 

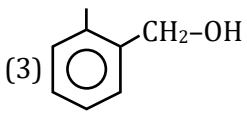
(4) 

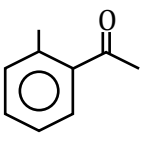
CAA305

21. Which does not give phthalic anhydride after heating with acidic $KMnO_4$?

(1) 

(2) 

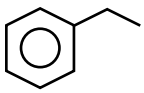
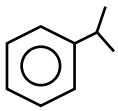
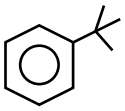
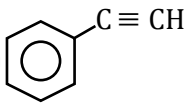
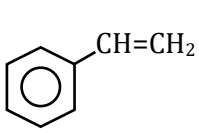
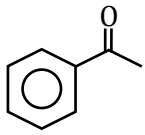
(3) 

(4) 

CAA306

AIM : FREE EDUCATION TO ALL APUL

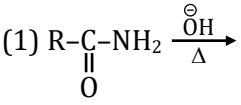
22. Which of the following give benzoic acid after heating with acidic $KMnO_4$?

(a) 
(b) 
(c) 
(d) 
(e) 
(f) 

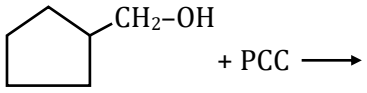
(1) a, b, d, e
(2) a, b, e
(3) a, b, d, e, f
(4) a, e, f

CAA307

23. Which reaction does not give carboxylic acid as a major product ?

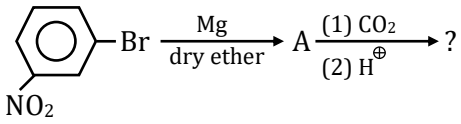
(1) 

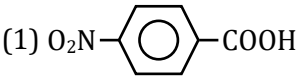
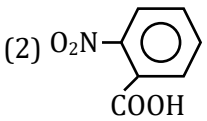
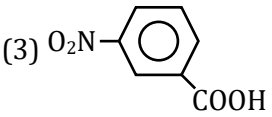
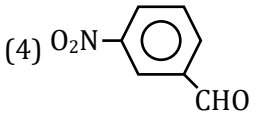
(2) $CH_3MgCl + CO_2 \longrightarrow \xrightarrow{H^{\oplus}}$

(3) 

(4) (1) and (3) both

CAA308

24. 

(1) 
(2) 
(3) 
(4) 

CAA309

25. **Statement-I :** Carboxylic acids exist as dimer in vapour or in aprotic solvents.
Statement-II : The hydrogen bonds are not broken completely even in the vapour phase.
(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

CAA310

26. **Assertion :** Carboxylate ion is more stable than phenoxide ion.
Reason : The negative charge is delocalized over two more EN oxygen atom.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CAA311

27. Which is most acidic ?
(1) $CF_3–COOH$
(2) $CCl_3–COOH$
(3) $CHCl_2–COOH$
(4) $O_2N–CH_2–COOH$

CAA312

28. **Assertion :** Direct attachment of groups such as phenyl or vinyl to the carboxylic acid, increases the acidity.
Reason : This is because of greater electronegativity of sp^2 hybridised carbon attached to carboxylic group.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CAA313

29. In which reaction amide is not obtained ?

(1) $CH_3–\underset{\underset{O}{\|}}{C}–OH + NH_3 \longrightarrow$

(2) $R–C\equiv N + \underset{\textbf{1 eq.}}{H_2O} \xrightarrow{\ominus OH}$

(3) $CH_3–\underset{\underset{O}{\|}}{C}–O–CH_3 + NH_3 \longrightarrow$

(4) All

CAA314

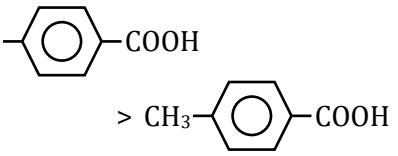
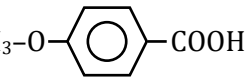
AIM : FREE EDUCATION TO ALL APUL

30. **Statement-I :** Diborane reduces carboxylic acid but not ester, nitro, halo etc.
Statement-II : $NaBH_4$ does not reduce the carboxylic acid.
(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

CAA315

31. In which reaction propane is not obtained ?
(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-COONa \xrightarrow[\Delta]{NaOH/CaO}$
(2) $CH_3-CH_2-CH_2-COONa \xrightarrow[\Delta]{NaOH/CaO}$
(3) $CH_3-CH_2-CH_2-COONa \xrightarrow{electrolysis}$
(4) All

CAA316

32. Which is not correct acidic order ?
(1) $CH_2F-COOH > CH_3-COOH$
(2) $FCH_2-CH_2-CH_2-COOH < CH_3-CHFCH_2-COOH$
(3)  >
(4)  > 

CAA317

33. Match the column-I with column-II

Column-I	Column-II
(1) $\begin{matrix}CH_3\\H\end{matrix}\!\!>\!C=O + H-CN$	(A) Acetal
(2) $\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!C=O + 2Et-OH$	(B) Ketal
(3) $\begin{matrix}CH_3\\H\end{matrix}\!\!>\!C=O + 2Et-OH$	(C) Cyanohydrine
(4) $CH_3-CH_2-CHO \xrightarrow{NaOH}$	(D) Aldol

(1) $1 \to C ; 2 \to A ; 3 \to B ; 4 \to D$
(2) $1 \to D ; 2 \to B ; 3 \to C ; 4 \to A$
(3) $1 \to C ; 2 \to B ; 3 \to A ; 4 \to D$
(4) $1 \to C ; 2 \to D ; 3 \to C ; 4 \to B$

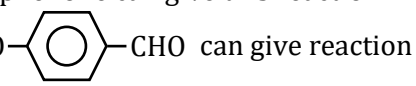
CAA318

34. Which gives Aldol condensation and which gives Cannizzaro reaction –
(a) 2-Methylpentanal
(b) 2,2-Dimethylbutanal
(c) Phenyl acetaldehyde
(d) Cyclohexanone
(e) 1-Phenyl propanone
(f) Benzophenone
(g) Benzaldehyde
Correct set is
(1) Aldol condensation – a, c, d, e
Cannizzaro – b, f, g
(2) Aldol condensation – a, c, e
Cannizzaro – b, f, g
(3) Aldol condensation – a, c, d, e
Cannizzaro – b, g
(4) Aldol condensation – a, c, e
Cannizzaro – b, g

CAA319

35. In aldol reaction of propanal and butanal, correct statement is -
(1) It is addition than elimination reaction.
(2) Propanal can act as nucleophile.
(3) Butanal can act as electrophile.
(4) All are correct

CAA320

36. Correct statement about Cannizzaro reaction is/are
(a) Benzophenone can give this reaction
(b)  can give reaction
(c) More reactive aldehyde is oxidised.
(d) Less reactive aldehyde is oxidised.
(e) It is overall disproportionation reaction
(1) a, b, c, e (2) b, d, e
(3) b, c, d, e (4) b, c, e

CAA321

37. **Assertion :** Ketone does not give Cannizzaro.
Reason : Ketone have α-H.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CAA322

38. An organic compound (A) having molecular formula $C_6H_{12}O_2$ gives an acid (B) and alcohol (C) on acidic hydrolysis. (C) is not oxidised by chromic acid. compound 'A' is

(1) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-O-CH_2-CH_2-CH_3$

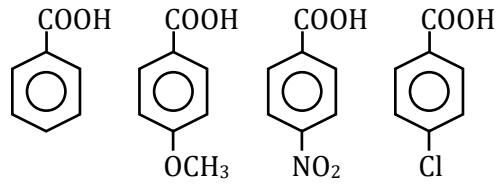
(2) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-O-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(3) $CH_3-\underset{\underset{O}{\|}}{C}-O-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(4) $CH_3-\underset{\underset{O}{\|}}{\overset{\overset{CH_3}{|}}{C}}-O-\underset{\underset{CH_3}{|}}{C}-CH_3$

CAA323

39. Acidic order of



(I) (II) (III) (IV)

(1) IV > III > I > II (2) III > IV > I > II

(3) III = IV > I > II (4) IV > III > II > I

CAA324

40. Match the column-I with Column-II

Column-I		Column-II (differentiated by)	
(A)	Acetophenone and Benzophenone	(P)	$NaHCO_3$
(B)	Phenol and Benzoic acid	(Q)	$NaHSO_3$
(C)	Propanone and Ethanal	(R)	I_2/NaOH
(D)	Acetone and acetophenone	(S)	Fehling solution

(1) A → P ; B → Q ; C → R ; D → S

(2) A → R ; B → P ; C → S ; D → Q

(3) A → R ; B → P ; C → Q ; D → S

(4) A → Q ; B → P ; C → S ; D → R

CAA325

41. **Statement-I :** Cyclohexanone forms cyanohydrin in good yield but 2, 2, 6-trimethyl cyclo hexanone does not.

Statement-II : 2, 2, 6-trimethyl cyclo hexanone has not α-H.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CAA326

42. **Assertion :** Out of two $-NH_2$ groups in semicarbazide, only one is involved in formation of semicarbazone.

Reason : One $-NH_2$ is involved in resonance which is poor nucleophile.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

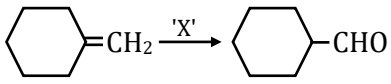
CAA327

43. H–COOH and CH_3–COOH can be differentiated by

(1) NaOH/I_2 (2) $NaHCO_3$

(3) Tollen's reagent (4) NaOH

CAA328

44. 

Reagent 'X' is

(1) (I) BH_3, (II) $H_2O_2/\overset{\ominus}{O}H$, (III) $H^{\oplus}/KMnO_4$

(2) (I) $H^{\oplus}/H_2O$, (II) PCC

(3) BH_3, $H_2O_2/\overset{\ominus}{O}H$, PCC

(4) $O_3/Zn/H_2O$, dil. NaOH

CAA329

45. **Statement-I :** Phenoxide ion has more number of resonating structures than carboxylate ion so it's more stable than carboxylate ion.

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

Statement-II : Phenoxide ion is more reactive than phenol towards electrophilic substitution.

(1) Both Statement I and Statement II are true.

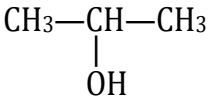
(2) Both Statement I and Statement II are false.

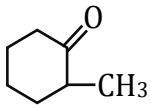
(3) Statement I is true but Statement II is false.

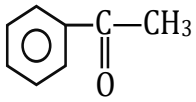
(4) Statement I is false but Statement II is true.

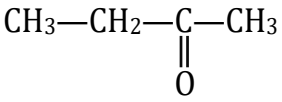
CAA330

46. Which does not give iodoform test ?

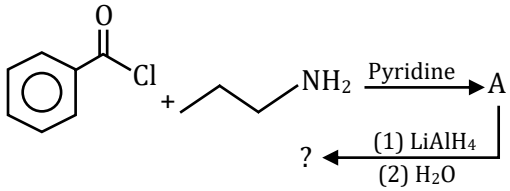
(1) 

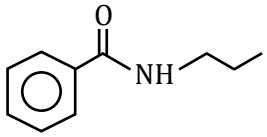
(2) 

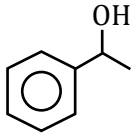
(3) 

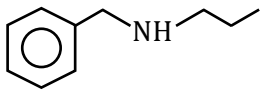
(4) 

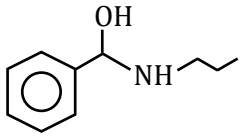
CAA331

47. 

(1) 

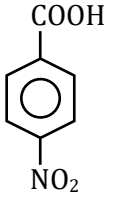
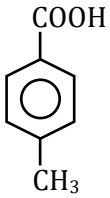
(2) 

(3) 

(4) 

CAA332

48. Assertion :

 is more acidic than 

Reason : –M of $-NO_2$ increases stability of carboxylate ion.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

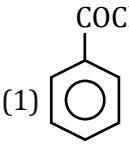
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

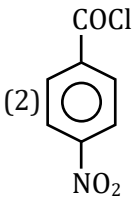
(3) Assertion is true but Reason is false.

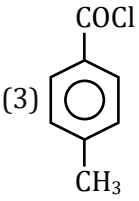
(4) Both Assertion and Reason are false.

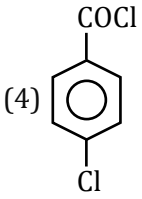
CAA333

49. Which is maximum reactive towards hydrolysis ?

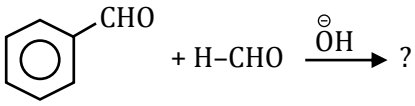
(1) 

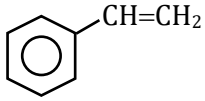
(2) 

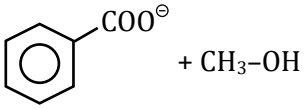
(3) 

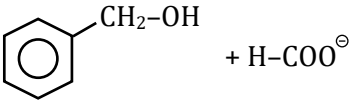
(4) 

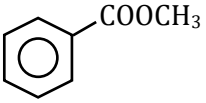
CAA334

50. 

(1) 

(2) 

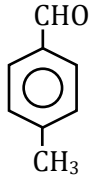
(3) 

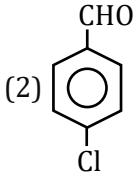
(4) 

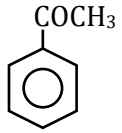
CAA335

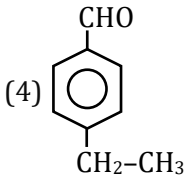
AIM : FREE EDUCATION TO ALL APUL

51. Which have maximum electrophilicity ?

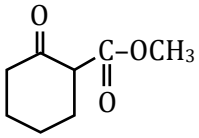
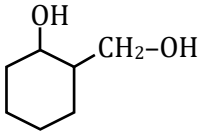
(1) 

(2) 

(3) 

(4) 

CAA336

52. **Statement-I :** Reduction of  with $NaBH_4$ gives 

Statement-II : Ester is more reactive than aldehyde towards nucleophile.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CAA337

53. Correct Match regarding reaction of aldehyde with reagent and product.

Column-I Reagent		Column-II Product	
(P)	NH_2-OH	(A)	Semicar-bazone
(Q)	NH_2-R	(B)	Oxime
(R)	$H_2N-\overset{\overset{O}{\|\|}}{C}-NH-NH_2$	(C)	Hydrazone
(S)	NH_2-NH_2	(D)	Schift base

(1) P → B ; Q → D ; R → A ; S → C

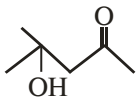
(2) P → B ; Q → D ; R → C ; S → A

(3) P → B ; Q → A ; R → D ; S → C

(4) P → C ; Q → D ; R → A ; S → B

CAA338

54. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A :  can be easily reduced using Zn-Hg/ HCl to 

Reason R : Zn-Hg/HCl is used to reduce carbonyl group to $-CH_2-$ group.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is false but R is true

(2) A is true but R is false

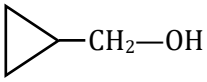
(3) Both A and R are true but R is not the correct explanation of A

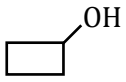
(4) Both A and R are true and R is the correct explanation of A

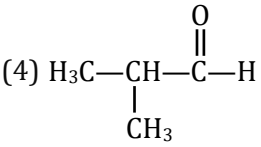
CAA339

55. Compound 'X' C_4H_8O which gives 2, 4-DNP derivative and negative iodoform test is :

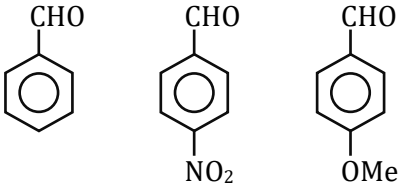
(1) 

(2) 

(3) 

(4) 

CAA340

56. 

I II III

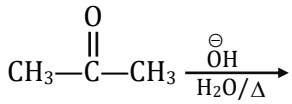
Which of the following orders is correct of the yield of products obtained when these compounds are treated with KCN/HCN?

(1) I > II > III (2) III > II > I

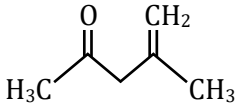
(3) II > I > III (4) III > I > II

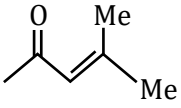
CAA341

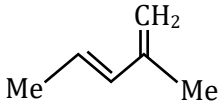
AIM : FREE EDUCATION TO ALL APUL

57. 

The major product which can be isolated from this reaction is:

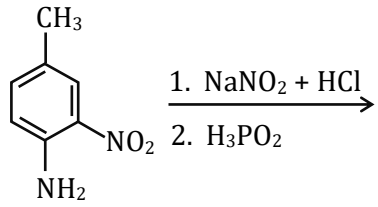
(1) 

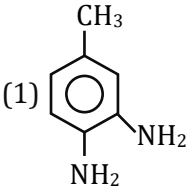
(2) 

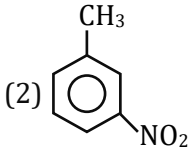
(3) 

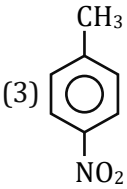
(4) None of these

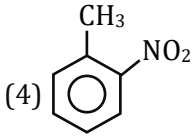
CAA342

58. 

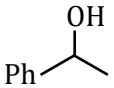
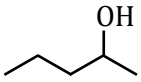
(1) 

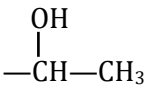
(2) 

(3) 

(4) 

CAA343

59. **(A)** →  and  can not be differentiated by iodoform reaction.

(R) → Alcohols having  group give haloform reaction.

(1) If both **(A)** and **(R)** are right and **(R)** is right explanation of **(A)**.

(2) If both **(A)** and **(R)** are right but **(R)** is not right explanation of **(A)**.

(3) If **(A)** is right but **(R)** is wrong.

(4) If both **(A)** and **(R)** are wrong.

CAA344

ANSWER KEY

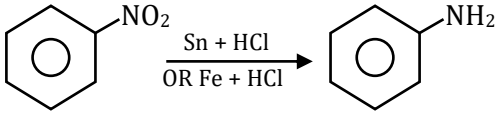
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	2	2	2	1	2	1	1	3	3	3	1	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	1	3	2	2	3	3	3	1	1	1	1	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	4	3	3	4	4	3	4	2	2	3	1	3	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	
Answer	2	3	1	2	3	2	2	1	1	4	3	2	2	1	

Amines

1. Which is/are correct statement -
(1) Adrenaline and ephedrine are used to increase blood pressure.
(2) Novocain is used as an anaesthetic.
(3) Benadryl is antihistamine
(4) All are correct

CAM203

2. **Assertion :-**



Reduction with iron scrap and hydrochloric acid is preferred.
Reason :- $FeCl_2$ formed gets hydrolysed to release hydrochloric acid.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CAM204

3. **Statement-I :-** Ammonolysis of halide has the disadvantage to primary amine as obtain good yield.
Statement-II :- Primary amine is obtained as a major product by taking large excess of ammonia.
(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

CAM205

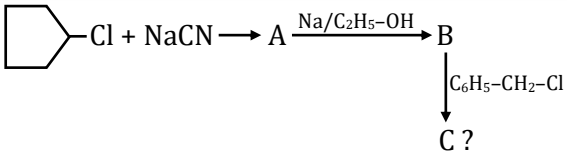
4. Which is most reactive towards amines.

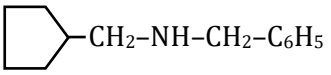
(1) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-I$ (2) $CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$

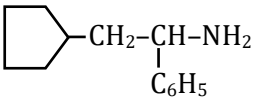
(3) $CH_3-\underset{\displaystyle Cl}{\underset{|}{CH}}-CH_3$ (4) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-F$

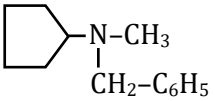
CAM206

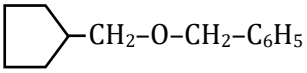
5.



(1) 

(2) 

(3) 

(4) 

CAM207

6. **Statement-I :** Reduction of ester / Acid / acid halide with $LiAlH_4$ gives Alcohol while amide gives amine.
Statement-II : $-NH_2$ has poor leaving tendency
(1) Both Statement-I and Statement-II are true and the Statement-II is a correct explanation of the Statement-I.
(2) Both Statement-I and Statement-II are true and but Statement-I is not a correct explanation of the Statement-I.
(3) Statement-I is true but Statement-II is false.
(4) Both Statement-I and Statement-II are false.

CAM208

7. **Assertion :** Primary amines have more boiling point than secondary and tertiary amines.
Reason : Primary amines have intermolecular association due to H-bonding.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

CAM209

AIM : FREE EDUCATION TO ALL APUL

8. Which have most solvation ?

(1) $R-\overset{\oplus}{N}H_3$ (2) $R_2\overset{\oplus}{N}H_2$

(3) $R_3\overset{\oplus}{N}H$ (4) Equal

CAM210

9. $CH_3-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow{NaOH} A \xrightarrow[1\ eqva]{Br_2} B$?

(1) $CH_3-\underset{\underset{O}{\|}}{C}-CH_2-Br$ (2) $CH_3-\underset{\underset{O}{\|}}{C}-Br$

(3) $Br-CH_2-\underset{\underset{O}{\|}}{C}-CH_2-Br$ (4) $Br-\underset{\underset{O}{\|}}{C}-Br$

CAM211

10. **Assertion :** The reaction of amine with acid halide is carried out in presence of base like pyridine.

Reason : Pyridine removes H-Cl formed I reaction so that equilibrium shift to right hand side.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CAM212

11. Incorrect statement about reaction of amine with HNO_2, is

(1) Aliphatic primary amine gives alcohol

(2) Secondary amines gives yellow oily liquid.

(3) Aromatic primary amine gives diazonium salt at room temperature.

(4) Tertiary amine forms salt.

CAM213

12. **Statement-I :-** Primary amine gives sulphonamide with Hinsberg reagent, which is soluble in alkali.

Statement-II :- Sulphonamide is strongly acidic due to presence of strong electron withdrawing sulphonamide group.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

CAM214

13. Correct statements is/are

(a) N, N-dimethyl benzene sulphonamide is insoluble in alkali.

(b) Tertiary amines do not react with benzene sulphonyl chloride.

(c) Benzenesulphonyl chloride is known as Hinsberg reagent.

(d) Primary amine gives alkali soluble product after reacting with Hinsberg reagent.

(1) a, c (2) a, c, d

(3) b, c, d (4) All are correct

CAM215

14. **Assertion :** Activating effect of $-NH_2$ in aniline is decreased due to acetylation.

Reason : Lone pair on nitrogen have resonance with oxygen in acetanilide so lone pair is less available for donation to benzene ring.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

CAM216

15. **Assertion :** Aniline does not give Friedel craft reaction.

Reason : In presence of Lewis acid ($AlCl_3$), aniline forms salt and nitrogen acquires positive charge which is strong deactivating group.

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true and but Reason is not a correct explanation of the Assertion.

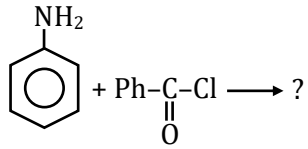
(3) Assertion is true but Reason is false.

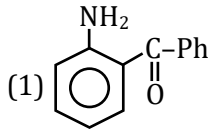
(4) Both Assertion and Reason are false.

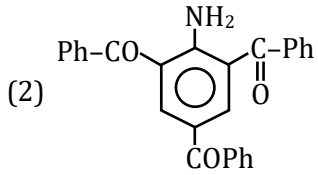
CAM217

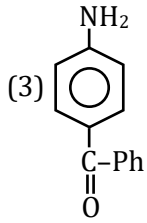
AIM : FREE EDUCATION TO ALL APUL

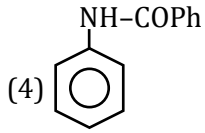
AIM : FREE EDUCATION TO ALL APUL

16. 

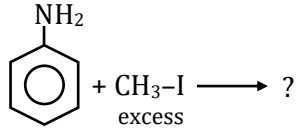
(1) 

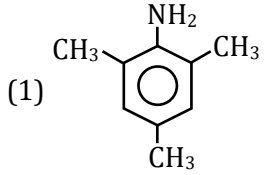
(2) 

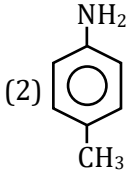
(3) 

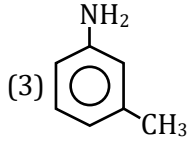
(4) 

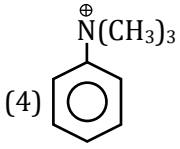
CAM218

17. 

(1) 

(2) 

(3) 

(4) 

CAM219

18. **Statement-I :-** The conversion of primary Aliphatic amine into diazonium salt is known as diazotization.

Statement-II :- Benzene diazonium salt is stable in cold or at 0°-5°C

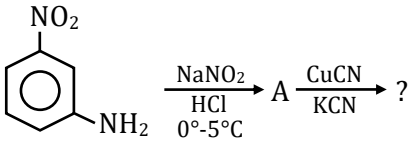
(1) Both Statement I and Statement II are true.

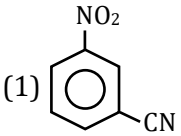
(2) Both Statement I and Statement II are false.

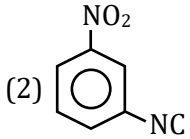
(3) Statement I is true but Statement II is false.

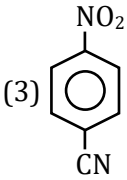
(4) Statement I is false but Statement II is true.

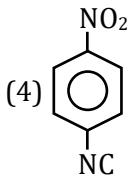
CAM220

19. 

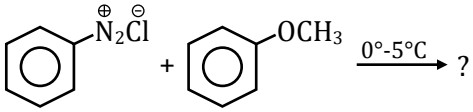
(1) 

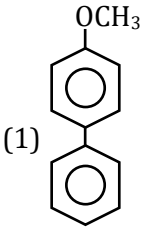
(2) 

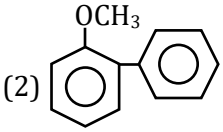
(3) 

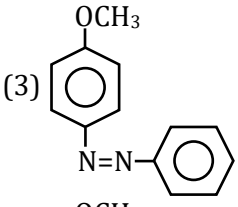
(4) 

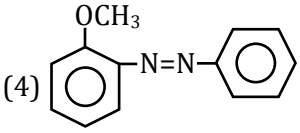
CAM221

20. 

(1) 

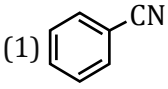
(2) 

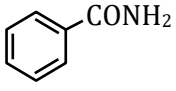
(3) 

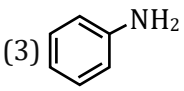
(4) 

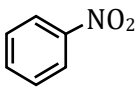
CAM222

21. A given nitrogen-containing aromatic compound (A) reacts with Sn/HCl, followed by HNO_2 to give an unstable compound (B). (B), on treatment with phenol, forms a beautiful coloured compound (C) with the molecular formula $C_{12}H_{10}N_2O$. The structure of compound (A) is :-

(1) 

(2) 

(3) 

(4) 

CAM223

22. Which of the following reactions is appropriate for converting acetamide to methanamine?

(1) Hoffmann hypobromamide reaction

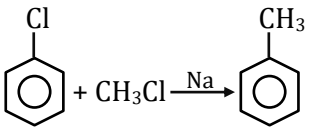
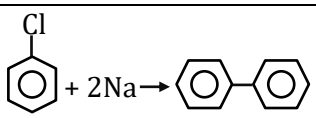
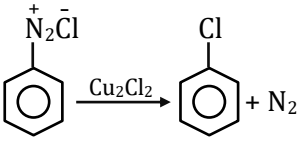
(2) Stephens reaction

(3) Gabriels phthalimide synthesis

(4) Carbylamine reaction

CAM224

23. Match **List I** with **List II**

	List-I		List-II	
A			I	Fittig reaction
B			II	Wurtz Fittig reaction
C			III	Finkelstein reaction
D	$C_2H_5Cl + NaI \rightarrow C_2H_5I + NaCl$		IV	Sandmeyer reaction

(1) A – II, B – I, C – III, D – IV

(2) A – III, B – II, C – IV, D – I

(3) A – IV, B – II, C – III, D – I

(4) A – II, B – I, C – IV, D – III

CAM225

24. $CH_3CH_2-C\equiv N \xrightarrow{X} CH_3CH_2-CHO$

The reagent/s X is (are):

(1) $SnCl_2/HCl$, H_2O/boil

(2) H_2/Pd-$BaSO_4$

(3) $LiAlH_4$/ether

(4) $NaBH_4$/ether, H_3O^+

CAM226

25. Which statement is NOT correct for p-toluenesulphonyl chloride ?

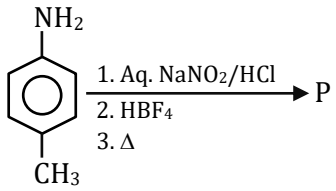
(1) It is known as Hinsberg's reagent.

(2) It is used to distinguish primary and secondary amines.

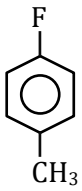
(3) On treatment with secondary amine, it leads to a product, that is soluble in alkali.

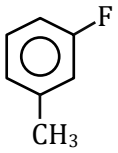
(4) All are correct

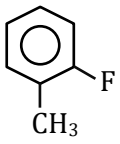
CAM227

26. 

P should be

(1) 

(2) 

(3) 

(4) A mixture of (1) and (3)

CAM228

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**1**	**1**	**1**	**1**	**2**	**1**	**1**	**1**	**3**	**1**	**4**	**1**	**1**
Question	16	17	18	19	20	21	22	23	24	25	26				
Answer	**4**	**4**	**4**	**1**	**3**	**4**	**1**	**4**	**1**	**3**	**1**				

AIM : FREE EDUCATION TO ALL APUL

Biomolecules

1. Which of the following polymer is stored in the liver of animals and is also known as animal starch ?
(1) Starch (2) Amylose
(3) Glycogen (4) Cellulose
CBM141

2. Which of the following base is not found in RNA ?
(1) Uracil
(2) Thymine
(3) Adenine
(4) Cytosine
CBM142

3. Which of the following vitamin is required to take regularly in diet ?
(1) Vit-A (2) Vit-D
(3) Vit-C (4) Vit-E
CBM143

4. Which of the following term is correct about enzyme ?
(1) Proteins
(2) Dinucleotides
(3) Biocatalysts
(4) Nucleic acids
CBM144

5. Sucrose is a :-
(1) Monosaccharide
(2) Disaccharide
(3) Reducing sugar
(4) Amino acid
CBM145

6. Which of the following is example of globular protein ?
(1) Keratin (2) Collagen
(3) Myosin (4) Insulin
CBM146

7. Which of the following polysaccharides structure is similar to amylopectin ?
(1) Starch (2) Cellulose
(3) Glycogen (4) None of these
CBM147

8. Which of the following B group vitamins can be stored in our body ?
(1) Vit-B_2 (2) Vit-B_6
(3) Vit-B_{12} (4) Vit-B_1
CBM148

9. Which of the following product is formed when D-glucose is treated with bromine water ?
(1) n-Hexane (2) Saccharic acid
(3) Gluconic acid (4) Oxalic acid
CBM149

10. Which of the following reactions of glucose can be explained only by its cyclic structure ?
(1) Glucose forms pentaacetate.
(2) Glucose is oxidised by nitric acid to gluconic acid.
(3) Glucose reacts with hydroxylamine to form an oxime.
(4) Pentaacetate of glucose does not react with hydroxyl amine.
CBM150

11. Which of the following statements is not true about glucose ?
(1) It is an aldohexose
(2) On heating with HI, it forms n-hexane.
(3) It is present in furanose form
(4) It does not give 2,4-DNP test
CBM151

12. Nucleic acids are the polymers of :-
(1) Nucleosides (2) Nucleotides
(3) Bases (4) Sugars
CBM152

13. Riboflavin is the chemical name of :-
(1) Vitamin-C (2) Vitamin-B_2
(3) Vitamin-B_6 (4) Vitamin-B_1
CBM153

14. Which of the following acid is a vitamin ?
(1) Aspartic acid
(2) Adipic acid
(3) Saccharic acid
(4) Ascorbic acid
CBM154

AIM : FREE EDUCATION TO ALL APUL

15. One molecule of sucrose on hydrolysis gives:-

(1) 2 molecules of fructose

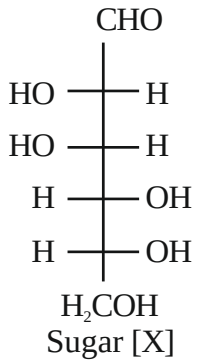
(2) 2 molecules of glucose

(3) 1 molecule of glucose + 1 molecule of fructose.

(4) 1 molecule of glucose + 1 molecule of maltose.

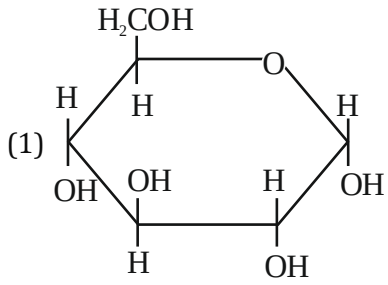
CBM155

16. The correct representation in six membered pyranose form for the following sugar [X] is

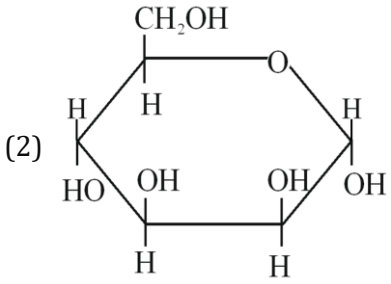


Sugar [X]

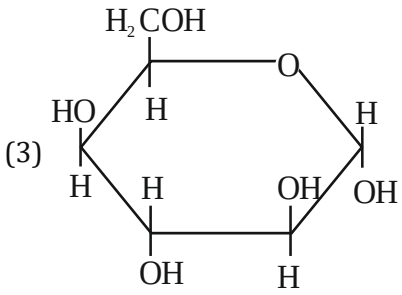
(1)



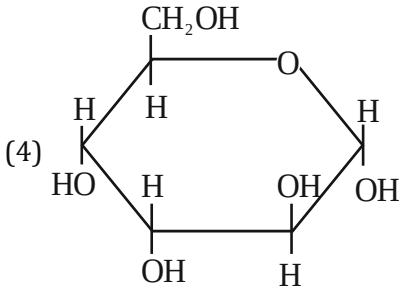
(2)



(3)



(4)



CBM156

17. Sulphur containing non-essential and essential amino acids are respectively :-

(1) Cystein and Histidine

(2) Histidine and Cystein

(3) Methionine and Cystein

(4) Cystein and Methionine

CBM157

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**3**	**1**	**2**	**4**	**3**	**3**	**3**	**4**	**3**	**2**	**2**	**4**	**3**
Question	16	17													
Answer	**2**	**4**													

AIM : FREE EDUCATION TO ALL APUL

Important Notes

AIM : FREE EDUCATION TO ALL ♥ APUL

Purification, Quantitative and Qualitative Analysis

1. **Statement-I :-** Crystallisation is one of the most commonly used techniques for the purification of liquid organic compounds.
Statement-II :- Crystallisation is based on the difference in the solubilites of the compound and the impurities in a suitable solvent.
(1) Both Statement I and Statement II are true.
(2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

CEC108

2. Distillation is used to Separate
(i) Volatile liquid from non volatile impurities
(ii) Liquid having sufficient difference in their boiling point.
(iii) Sublimable compounds from non sublimable compounds
(iv) Compounds takes place based on difference in solubility.
True statements are-
(1) i, ii, iii, iv (2) i, ii, iii only
(3) i, ii only (4) I, iii only

CEC109

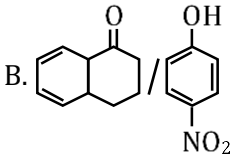
3. Match the column :-

Column-I		Column-II	
(A)	Fractional distillation	(P)	Liquid decompose at or below their B.P.
(B)	Distillation under Reduced pressure	(Q)	Substances which are steam volatile and are immisible with water
(C)	Steam distillation	(R)	Difference in B.P. of two liquids is not much
(D)	Differential extraction	(S)	Organic solvent and aqueous solution should be immisible with each other

(1) A → R ; B → P ; C → Q ; D → S
(2) A → R ; B → Q ; C → P ; D → S
(3) A → Q ; B → R ; C → P ; D → S
(4) A → P ; B → R ; C → S ; D → Q

CEC110

4. Match items of column I and II

Column-I (Mixture of compounds)	Column-II (Separation Technique)
A. H_2O/CH_2Cl_2	i. Crystallization
B. 	ii. Differential solvent extraction
C. Kerosene / Naphthalene	iii. Column chromatography
D. $C_6H_{12}O_6$/NaCl	iv. Fractional distillation

Correct match is :
(1) A-(iii), B-(iv), C-(ii), D-(i)
(2) A-(i), B-(iii), C-(ii), D-(iv)
(3) A-(ii), B-(iii), C-(iv), D-(i)
(4) A-(ii), B-(iv), C-(i), D-(iii)

CEC111

5. Match List I with List II:

List-I (Mixture)		List-II (Separation Technique)	
(A)	$CHCl_3 + C_6H_5NH_2$	I	Steam distillation
(B)	$C_6H_{14} + C_5H_{12}$	II	Differential extraction
(C)	$C_6H_5NH_2 + H_2O$	III	Distillation
(D)	Organic compound in H_2O	IV	Fractional distillation

(1) A-IV, B-I, C-III, D-II
(2) A-III, B-IV, C-I, D-II
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-IV, D-II

CEC112

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

6. Given below are two statements :

Statement I : In 'Lassaigne's Test, when both nitrogen and sulphur are present in an organic compound, sodium thiocyanate is formed.

Statement II : If both nitrogen and sulphur are present in an organic compound, then the excess of sodium used in sodium fusion will decompose the sodium thiocyanate formed to give NaCN and Na_2S.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

CEC113

7. Match the column :-

Column-I (Compound)		**Column-II (Colour for indication)**	
(i)	Nitrogen	(a)	Prussian Blue ppt
(ii)	Sulphur	(b)	Black ppt
(iii)	Chlorine	(c)	Yellow ppt
(iv)	Phosphorous	(d)	White ppt

(1) i → a ; ii → b ; iii → d ; iv → c

(2) i → a ; ii → b ; iii → c ; iv → d

(3) i → c ; ii → a ; iii → d ; iv → b

(4) i → d ; ii → c ; iii → b ; iv → a

CEC114

8. Match the column :-

Column-I		**Column-II**	
(i)	$Fe_4[Fe(CN)_6]_3.xH_2O$	(P)	Yellow ppt
(ii)	$[Fe(CN)_5NOS]^{4-}$	(Q)	Blood red
(iii)	$(NH_4)_3.PO_4.12MoO_3$	(R)	Prussian blue
(iv)	$[Fe(SCN)]^{2+}$	(S)	Violet

(1) i → R ; ii → S ; iii → P ; iv → Q

(2) i → R ; ii → P ; iii → S ; iv → Q

(3) i → S ; ii → R ; iii → P ; iv → Q

(4) i → P ; ii → S ; iii → R ; iv → Q

CEC115

9. In given question total correct Statements are:

(i) Dumas method is used for estimation of nitrogen.

(ii) Kjeldahl's method is used for estimation of nitrogen.

(iii) Carius method is used for Halogen only

(iv) I_2O_5 is used during estimation of phosphorus.

(v) $(NH_4)_3.PO_4.12MoO_3$ is used for estimation of phosphorous

(1) 4 (2) 3 (3) 2 (4) 1

CEC116

10. R_f value (Retardation factor)

(i) Distance moved by substance from base line (a)

(ii) Distance moved by solvent from base line (b)

(1) $\frac{a}{b}$ (2) $\frac{b}{a}$

(3) Both (4) None

CEC117

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**4**	**3**	**1**	**3**	**2**	**1**	**1**	**1**	**2**	**1**

Important Notes

AIM : FREE EDUCATION TO ALL APUL